

Emergency Core Cooling System Performance Criteria for Multi-Layered Silicon Carbide Nuclear Fuel Cladding

Robby Christian¹, Youho Lee², Hyun Gook Kang³

¹ Idaho National Laboratory, Idaho Falls, ID 83415, USA

² Department of Nuclear Engineering, Seoul National University, Seoul 08826, Republic of Korea

³ Mechanical, Aerospace, and Nuclear Engineering Department, Rensselaer Polytechnic Institute, Troy, NY 12180, USA

Corresponding author: kangh6@rpi.edu

Abstract

Multi-layered Silicon Carbide (SiC) cladding has been proposed to replace the current Zircaloy (Zr) cladding in existing nuclear power plants. Due to its ceramic property, SiC may have a different failure mechanism from that of metallic Zr clad. This characteristic should be considered in designing the plant's safety system performance, such that the fuel's integrity is reasonably assured during accident scenarios.

This paper aims at optimizing the Emergency Core Cooling System (ECCS) performance criteria in mitigating LBLOCA accidents at a Pressurized Water Reactor (PWR) using the uranium dioxide fuel and duplex or triplex SiC cladding structures. Clad failures by brittle fractures and thermal degradation were modeled in a fuel analysis code. This fuel model was coupled with the RELAP5 plant model by adapting the fuel's mesh configuration and formulating the effective heat transfer coefficients. A model to quantify the likelihood of clad functional failure as a combination of individual layer fractures at various axial positions was proposed. Updated SiC material properties were gathered from recent reference studies. Clad reliability estimates using the proposed model were compared with the estimates obtained using other models. The results were used to relax the criterion of required actuation timing of ECCS, such that the risk of the new plant design is less or equal to that of the initial plant configuration.

Triplex clad was more reliable than duplex clad and Zr, reducing the conditional core damage probability by three orders of magnitude in the reference LBLOCA scenario. Compressive stresses due to irradiation swelling played the determining role in the vulnerabilities of each clad design. The duplex clad was vulnerable to pressure-loading stresses while triplex clad was prone to fail due to thermal shocks experienced during the LOCA reflood phase. Additionally, it was concluded that the most vulnerable axial fuel rod location in the reference LBLOCA scenario was located at the highest axial power peaking location.

A sensitivity study on the ECCS timing density was conducted. The results showed that ECCS actuation timing could be extended from the initial technical specification, without increasing the fuel damage likelihood. The time extension was distributed in the order of several hundreds of seconds, which was comparable with time extensions offered by FeCrAl and Cr-coated-Zr clad designs in LBLOCA scenarios. This time extension may potentially reduce the failure probabilities of EDGs and pumps, hence improving the plant safety further.

Keywords: ECCS, SiC, LBLOCA

1. Introduction

Following the Fukushima-Daiichi accident, there has been significant research to replace current UO₂-Zr fuel design in existing LWR fleet with Accident Tolerant Fuels (ATFs) [1]. One of the best ATF candidates is Silicon Carbide (SiC) cladding as a substitute to Zr cladding [2]. SiC has a higher decomposition temperature and better oxidation resistance compared to Zr [3]. However, being a ceramic material, SiC is susceptible to brittle fractures which can cause the cladding structure to fail stochastically [4]. To overcome this limitation, researchers proposed multi-layered SiC clad structures [5]. The clad design includes a duplex structure with an inner Chemical Vapor Infiltration (CVI) SiC/SiC matrix composite layer and an outer Chemical Vapor Deposition (CVD) SiC layer, and a triplex structure with a CVI SiC/SiC matrix composite layer sandwiched between inner and outer CVD SiC layers as illustrated in Figure 1.

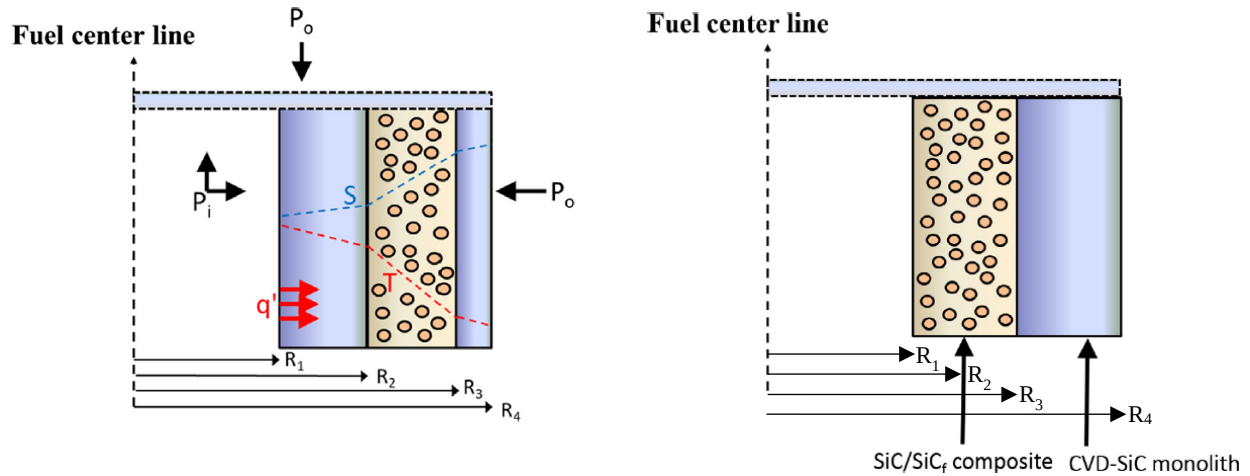


Figure 1. Triplex (left) and duplex (right) SiC cladding structure [6]

Formulations to calculate stresses in SiC multi-layered structures and the corresponding clad fracture likelihood have been detailed by Lee [6], [7]. This stress model has been adopted in other studies. Zhang [8] used this stress model and modified the statistical Weibull fracture model to estimate cladding failure in Pressurized Water Reactor (PWR) operation condition. Deng [9] implemented this stress model in an established fuel assessment software to estimate the full-core response of the duplex SiC cladding structure. Meanwhile, Singh [10] took a different approach by utilizing a finite-element-based software to assess the response of a single SiC/SiC matrix composite layer instead. The finite-element approach was also used in other research [11], [12] using commercial software.

While previous research has laid out the foundation to estimate SiC cladding reliability, the analysis was mainly based on steady-state reactor conditions. There is a lack of research to determine SiC clad integrity estimates in varying accident conditions. Such research may not be required if the ATF is truly accident-tolerant, i.e., does not fail in any accident scenario. However, a trade-off study [2] has revealed that no such ideal ATF candidate exists. Each ATF candidate has its benefits and drawbacks. Therefore, the accident-tolerance feature of a nuclear plant should be optimized by properly adapting the accident mitigation actions to the fuel's characteristics.

SiC is susceptible to fractures caused by thermal quenching events [4]. Such event occurs when core is reflooded after uncover, which happen during the initiation of Emergency Core Cooling System (ECCS) following a Large Break Loss of Coolant Accident (LBLOCA). This study formulated the framework in designing the ECCS performance for the uranium dioxide (UO_2) fuel with a duplex or triplex SiC clad in a PWR reactor. In doing so, the cladding's reliability in various Large Break Loss of Coolant Accident (LBLOCA) scenarios was investigated. To estimate the clad's reliability, the cladding functional failure was defined as a combination of individual layer's failure at different axial heights. Based on this clad reliability profile, the ECCS performance was optimized on the basis of probabilistic risk.

2. Methodologies

2.1. ECCS Performance Criteria

ECCS criteria for current uranium oxide fuels with Zircaloy (Zr) cladding could be estimated deterministically based on the conservative limits of Peak Cladding Temperature (PCT), Equivalent Cladding Reacted (ECR), and hydrogen generation [13]. The reactor core is in a safe state when ECCS operation successfully limits these parameters, i.e. scenario number 1 in Figure 2. On the contrary, scenario number 1 with SiC cladding may not fully guarantee the core is safe due to SiC's stochasticity of brittle fractures. For that reason, ECCS performance should be designed to limit the core damage likelihood in scenario number 1 and number 2, i.e. when ECCS fail to actuate. This study focuses on the timing aspect of ECCS actuation, as it is considered important in quantifying the safety benefits of ATFs [14], [15].

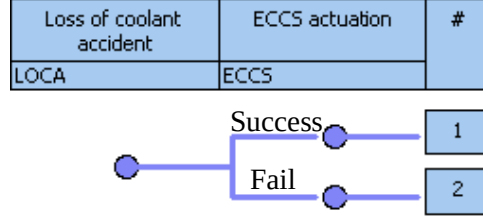


Figure 2. Illustration of ECCS actuation logic

Variations of ECCS actuation timing and the corresponding state of the fuel rod are illustrated in Figure 3. Following a Large Break LOCA accident, ECCS is typically activated within ~30 seconds to reflood the core [16], [17]. If ECCS starts too late, the fuel cladding fails deterministically due to its PCT exceeding the failure threshold or over-oxidation [18]. This event corresponds to scenario number 2 in Figure 2, while earlier events are in scenario number 1. This PCT threshold for Zr is 1204 °C [19] while for SiC is its thermal decomposition temperature of 2730 °C [7]. With such significant PCT difference, scenario number 2 with SiC clad is less likely to happen compared to Zr. However, SiC brittle fracture may cause a core damage as well in scenario number 1. With these considerations, the conditional core damage probability (P_{CD}) was formulated as an integral of clad failure events (P_{clad}) throughout the ECCS possible range of operation (P_{ECCS}) as shown in Equation (1).

$$P_{CD} = \int_{t=0}^{\infty} P_{ECCS} \times P_{clad} dt \quad (1)$$

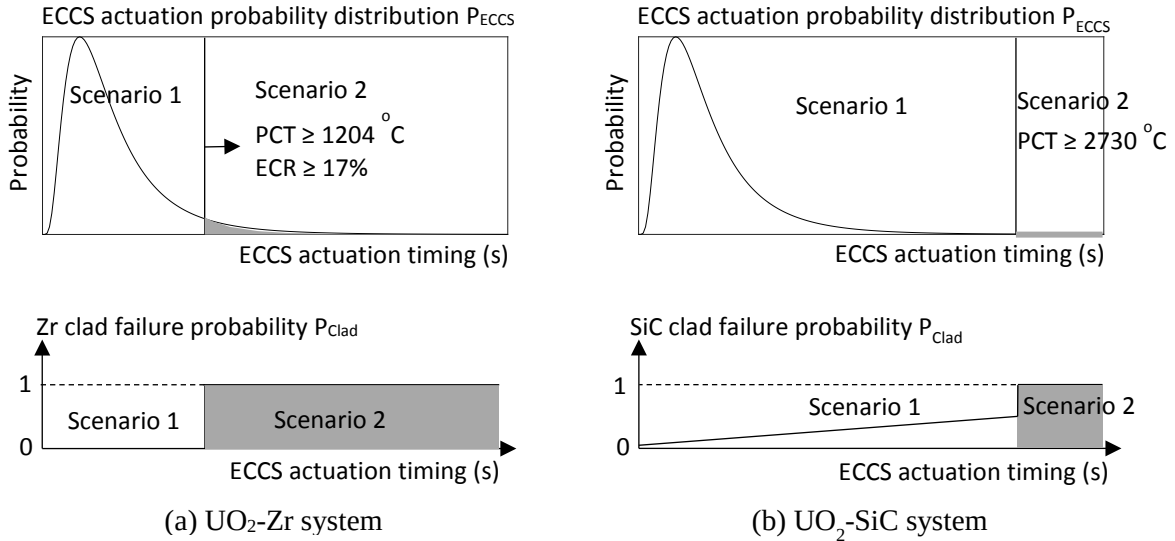


Figure 3. Illustration on uncertainties of ECCS actuation timing and clad reliability

The modification of existing nuclear power plants by adopting ATF fuel and adjusting ECCS performance should be justified to obtain the regulatory approval. This justification, as stated in reference [20], was done by ensuring that the risk increase due to the modification is reasonably small or negative. For that reason, it was deemed appropriate to set the ECCS design criterion such that the modified plant's risk is less than the initial plant risk, as shown in Equation (2). To obtain the bounding P_{ECCS} density parameters that satisfy Equation (2), an optimization process was done using Equation (3).

$$P_{ECCS} | (P_{CD-SiC} \leq P_{CD-Zr}) \quad (2)$$

$$\arg \min_{P_{ECCS-SiC}} \left[\left(\int P_{ECCS} \times P_{clad} \right)_{SiC} - \left(\int P_{ECCS} \times P_{clad} \right)_{Zr} \right]^2 \quad (3)$$

2.2. Temperature and stress distribution in the fuel rod

The radial temperature distribution in the fuel rod during a steady-state operation is given in reference [6]. The temperature of the cladding gap (T_{gap}) was determined by averaging all temperature data along the fuel's outer surface and the cladding's inner surface. This averaged gap temperature was used to determine the plenum pressure (P_{gap}) by assuming the ideal gas law. Data on the fuel rod used in this study is given in Table 1.

Table 1. Fuel rod dimensions

Parameter	Value
Height (active fuel portion)	3.81 m
Fuel pellet radius	4.1275 mm
Layer 1 inner radius (R1)	4.2164 mm
Clad outer radius (R4)	4.8514 mm
Triplex clad	
Layer 2 inner radius (R2)	4.5022 mm
Layer 3 inner radius (R3)	4.7879 mm
Duplex clad	
Layer 2 inner radius (R2)	4.3752 mm
Layer 3 inner radius (R3)	4.5339 mm

Radial temperature distribution during transients was solved with the implicit finite difference method. Heat transfer in the axial axis during reactor transients was omitted because it was assumed negligible due to the clad's thin cylindrical geometry. The mesh configuration for the triplex cladding structure is shown in Figure 4. Heat transfer direction was assumed as coming into a mesh node from neighboring nodes. The heat conductivity k , volumetric heat capacity ρC_p and linear heat generation q' anywhere between mesh nodes were assumed spatially constant. Their values were temperature-dependent, which was taken as the mean temperature between the nodes. These are the same assumptions used in the RELAP5 reactor safety analysis code [21].

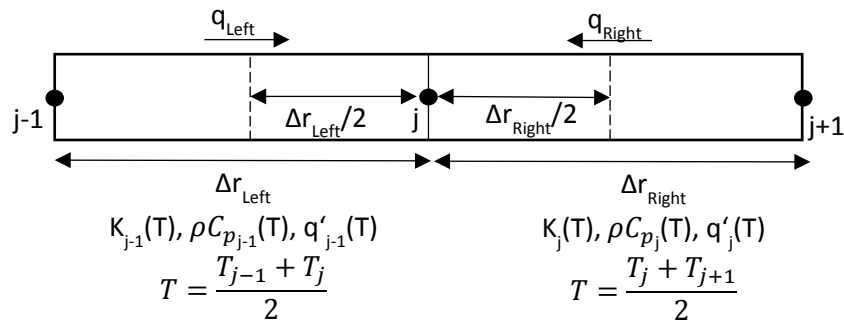


Figure 4. Radial mesh arrangement

$$X = [X_{f[1]} \quad X_{f[j]} \quad \dots \quad X_{fs} \quad X_{c[1][1]} \quad X_{c[i][j]} \quad \dots \quad X_{c[3][s]}]^T \quad X = A^{-1}b \quad (4)$$

$$A = \begin{bmatrix} A_{f[1]}(1) & A_{f[1]}(2) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & A_{f[j]}(1) & A_{f[j]}(2) & A_{f[j]}(3) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & 0 & 0 & 0 & 0 \\ 0 & \dots & 0 & A_{fs}(1) & A_{fs}(2) & A_{fs}(3) & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & A_{c[1][1]}(1) & A_{c[1][1]}(2) & A_{c[1][1]}(3) & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & A_{c[i][j]}(1) & A_{c[i][j]}(2) & A_{c[i][j]}(3) & \dots \\ 0 & 0 & 0 & 0 & \dots & 0 & \ddots & \ddots & \ddots \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & A_{c[3]s}(1) & A_{c[3]s}(2) \end{bmatrix}$$

$$b = [b_{f[1]} \quad b_{f[i]} \quad \dots \quad b_{fs} \quad b_{c[1][1]} \quad b_{c[i][j]} \quad \dots \quad b_{c[3]s}]^T$$

Transient temperature distribution in the radial direction was solved using Equation (4) with a sparse matrix solver. The terms in the equation represent nodal parameters from various radial positions in the fuel rod. Detailed derivations of these parameters are available in reference [7], however, the resulting equations were adapted in this section to reflect the aforementioned mesh arrangements shown in Figure 4. Parameters for the fuel center node ($r=0$) are:

$$A_{f[1]} = \left[- \left(\frac{k_{f[1]}}{\Delta r_f} A_{[1]} + \frac{(\rho C_p)_{f[1]}}{\Delta t} V_{[1]} \right) \frac{k_{f[1]}}{\Delta r_f} A_{[1]} \right]$$

$$A_{[1]} = \pi \Delta r_f, \quad V_{[1]} = \pi \left(\frac{\Delta r_f}{2} \right)^2$$

$$X_{f[1]} = [T_1^{t+1} \quad T_2^{t+1}]^T, \quad b_{f[1]} = - \frac{(\rho C_p)_{f[1]} V_{[1]} T_1^t}{\Delta t} - q''' P(t) V_{[1]}$$

For the fuel meat nodes ($0 < r_{[j]} < R_f$):

$$A_{f[j]} = \left[\frac{k_{f[j-1]}}{\Delta r_f} A_{[j-1]} - \left(\frac{k_{f[j-1]} A_{[j-1]}}{\Delta r_f} + \frac{k_{f[j]} A_{[j]}}{\Delta r_f} + \frac{(\rho C_p)_{f[j-1]} V_{[j-1]} + (\rho C_p)_{f[j]} V_{[j]}}{\Delta t} \right) \frac{k_{f[j]} A_{[j]}}{\Delta r_f} \right]$$

$$A_{[j-1]} = 2\pi \left(r_{[j]} - \frac{\Delta r_f}{2} \right), \quad A_{[j]} = 2\pi \left(r_{[j]} + \frac{\Delta r_f}{2} \right)$$

$$V_{[j-1]} = \pi \Delta r_f \left(r_{[j]} - \frac{\Delta r_f}{4} \right), \quad V_{[j]} = \pi \Delta r_f \left(r_{[j]} + \frac{\Delta r_f}{4} \right)$$

$$X_{f[j]} = [T_{j-1}^{t+1} \quad T_j^{t+1} \quad T_{j+1}^{t+1}]^T$$

$$b_{f[j]} = - \frac{((\rho C_p)_{f[j-1]} V_{[j-1]} + (\rho C_p)_{f[j]} V_{[j]}) T_j^t}{\Delta t} - q''' P(t) (V_{[j-1]} + V_{[j]})$$

For the fuel outer node ($r_{[j]}=R_f$):

$$A_{fs} = \left[\frac{k_{f[j-1]}}{\Delta r_f} A_{[j-1]} - \left(\frac{k_{f[j-1]} A_{[j-1]}}{\Delta r_f} + h_{gap} A_{[j]} + \frac{(\rho C_p)_{f[j-1]} V_{[j-1]}}{\Delta t} \right) h_{gap} A_{[j]} \right]$$

$$A_{[j-1]} = 2\pi \left(R_f - \frac{\Delta r_f}{2} \right), \quad A_{[j]} = 2\pi R_f$$

$$V_{[j-1]} = \pi \Delta r_f \left(R_f - \frac{\Delta r_f}{4} \right)$$

$$X_{fs} = [T_{[j-1]}^{t+1} \quad T_{[j]}^{t+1} \quad T_{[j+1]}^{t+1}]^T$$

$$b_{fs} = - \frac{(\rho C_p)_{f[j-1]} V_{[j-1]} T_{[j]}^t}{\Delta t} - q''' P(t) V_{[j-1]}$$

For the clad inner surface node ($r_{[j]}=R_1$):

$$\begin{aligned}
A_{c[1][1]} &= \left[h_{gap} A_{[j-1]} - \left(\frac{k_1 A_{[j-1]}}{\Delta r_1} + \frac{k_1 A_{[j]}}{\Delta r_1} + \frac{(\rho C_p)_{[i][j]} V_{[j]}}{\Delta t} \right) \frac{k_1 A_{[j]}}{\Delta r_1} \right] \\
A_{[j-1]} &= 2\pi R_f, \quad A_{[j]} = 2\pi \left(R_1 + \frac{\Delta r_1}{2} \right) \\
V_{[j]} &= \pi \Delta r_1 \left(R_1 + \frac{\Delta r_1}{4} \right) \\
X_{c[1][1]} &= [T_{[j-1]}^{t+1} \quad T_{[j]}^{t+1} \quad T_{[j+1]}^{t+1}]^T \\
b_{c[1][1]} &= - \frac{(\rho C_p)_{c[j]} V_{[j]} T_{[j]}^t}{\Delta t}
\end{aligned} \tag{8}$$

For the i-th layer clad nodes:

$$\begin{aligned}
A_{c[i][j]} &= \left[\frac{k_{[i]} A_{[j-1]}}{\Delta r_{[i]}} - \left(\frac{k_{[i]} A_{[j-1]}}{\Delta r_{[i]}} + \frac{k_{[i]} A_{[j]}}{\Delta r_{[i]}} + \frac{(\rho C_p V)_{[i][j-1]} + (\rho C_p V)_{[i][j]}}{\Delta t} \right) \frac{k_{[i]} A_{[j]}}{\Delta r_{[i]}} \right] \\
A_{[j-1]} &= 2\pi \left(r_{[j]} - \frac{\Delta r_{[i]}}{2} \right), \quad A_{[j]} = 2\pi \left(r_{[j]} + \frac{\Delta r_{[i]}}{2} \right) \\
V_{[j-1]} &= \pi \Delta r_{[i]} \left(r_{[j]} - \frac{\Delta r_{[i]}}{4} \right), \quad V_{[j]} = \pi \Delta r_{[i]} \left(r_{[j]} + \frac{\Delta r_{[i]}}{4} \right) \\
X_{c[i][j]} &= [T_{[j-1]}^{t+1} \quad T_{[j]}^{t+1} \quad T_{[j+1]}^{t+1}]^T \\
b_{c[i][j]} &= - \frac{(\rho C_p)_{c[i]} (V_{[j-1]} + V_{[j]}) T_j^t}{\Delta t}
\end{aligned} \tag{9}$$

For the interface node between clad Layer i and Layer i+1:

$$\begin{aligned}
A_{c[i]s} &= \left[\frac{k_{[i]} A_{[j-1]}}{\Delta r_{[i]}} - \left(\frac{k_{[i]} A_{[j-1]}}{\Delta r_{[i]}} + \frac{k_{[i+1]} A_{[j]}}{\Delta r_{[i+1]}} + \frac{(\rho C_p)_{[i]} V_{[j-1]} + (\rho C_p)_{[i]} V_{[j]}}{\Delta t} \right) \frac{k_{[i+1]} A_{[j]}}{\Delta r_{[i+1]}} \right] \\
A_{[j-1]} &= 2\pi \left(R_{[i+1]} - \frac{\Delta r_{[i]}}{2} \right), \quad A_{[j]} = 2\pi \left(R_{[i+1]} + \frac{\Delta r_{[i+1]}}{2} \right) \\
V_{[j-1]} &= \pi \Delta r_{[i]} \left(R_{[i+1]} - \frac{\Delta r_{[i]}}{4} \right), \quad V_{[j]} = \pi \Delta r_{[i+1]} \left(R_{[i+1]} + \frac{\Delta r_{[i+1]}}{4} \right) \\
X_{c1s} &= [T_{[j-1]}^{t+1} \quad T_{[j]}^{t+1} \quad T_{[j+1]}^{t+1}]^T \\
b_{c1s} &= - \frac{(\rho C_p)_{[i]} V_{[j-1]} + (\rho C_p)_{[i]} V_{[j]}}{\Delta t} T_{[j]}^t
\end{aligned} \tag{10}$$

For the clad outermost surface node ($r_i=R_4$):

$$\begin{aligned}
A_{c[3]s} &= \left[\frac{k_3 A_{[j-1]}}{\Delta r_3} - \left(\frac{k_3 A_{[j-1]}}{\Delta r_3} + h_{conv} A_{[j]} + \frac{(\rho C_p)_3 V_{[j-1]}}{\Delta t} \right) \right] \\
A_{[j-1]} &= 2\pi \left(R_4 - \frac{\Delta r_3}{2} \right), \quad A_{[j]} = 2\pi R_4 \\
V_{[j-1]} &= \pi \Delta r_3 \left(R_4 - \frac{\Delta r_3}{4} \right) \\
X_{c[3]s} &= [T_{[j-1]}^{t+1} \quad T_{[j]}^{t+1}]^T \\
b_{c[3]s} &= - \frac{(\rho C_p)_3 V_{[j-1]} T_{[j]}^t}{\Delta t} - h_{conv} A_{[j]} T_{\sim}
\end{aligned} \tag{11}$$

Where Δr is the radial distance between mesh nodes, q''' is the volumetric heat generation rate, $P(t)$ is the ratio of reactor power at time t to the steady-state power, h_{gap} is the gap conductance, h_{conv} is the convective heat transfer coefficient at the outer cladding surface, and $T_{\sim}$ is the bulk coolant temperature.

Formulations to calculate cladding stress were adopted from reference [6]. This adoption encompasses the underlying equations (e.g., Lamé equations, Hooke's Law), assumptions (e.g., plane-strain assumption, isotropic elasticity) and boundary conditions (e.g., continuity at layer boundaries, constant axial strains). Three types of stresses were calculated, i.e., mechanical stresses, thermal stresses, and irradiation swelling stresses. Mechanical stresses are caused by the pressure differences between the fuel plenum and coolant. Thermal stresses are induced by the varying extent of thermal expansion in the cladding layers. Irradiation swelling stresses are generated by the buildup of material damage due to neutron irradiation during normal reactor operations [22], [23]. The cladding stresses propagate in the radial, hoop and axial principal directions. These stresses may cause fractures as explained in the following section.

2.3. SiC cladding brittle fracture

2.3.1. Brittle fracture modeling

SiC is a brittle ceramic material, as opposed to the ductile metallic Zircaloy cladding. When SiC is under tension, the stresses concentrate locally around pre-existing material flaws as illustrated in Figure 5. If there is a sufficiently large flaw, cracks can occur originating from the flaw's location. As material bonds are broken by the new cracks, the material loses strength even further and consequently fractures if the stress remains. Because of the uncertainty of these flaws, the fracture likelihood is characterized with an empirical two-parameter Weibull distribution [6], [7], [9], [24]. This distribution is based on the weakest-link theory, in which the failure of a component (e.g., a chain) is dictated by the failure of a single constitutive element (e.g., a link) [24]. This theory agrees with the aforementioned qualitative understanding of ceramic fracture mechanism.

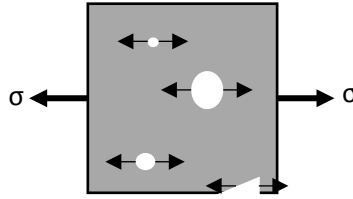


Figure 5. Local stress concentration in ceramic materials

The Weibull reliability function for each layer of SiC cladding is given by the general form in Equation (12) [6], where Weibull parameters are σ_0 and m while σ_r , σ_θ and σ_z indicate radial, hoop and tensile stresses respectively. Weibull parameters are empirically fitted from tensile stress experiments on specimens with volume V_0 . These parameters are applied to extrapolate survival likelihood of samples with a different volume V , assuming that the clad flaws are randomly distributed. In this study, the unit volume V was taken as the clad volume at each axial height and each layer. Clad fracture probability is simply the complement of Weibull reliability function as shown in Equation (13).

$$P_s(V) = \exp \left[-\frac{1}{V_0} \int_V \left(\frac{\sigma_r(\vec{r})}{\sigma_0} \right)^m + \left(\frac{\sigma_\theta(\vec{r})}{\sigma_0} \right)^m + \left(\frac{\sigma_z(\vec{r})}{\sigma_0} \right)^m dV \right] \quad (12)$$

$$P_f(V) = 1 - P_s(V) \quad (13)$$

$$P_f(V, t) | 0 \leq t \leq t_1 = P_f(V, t) (\max(\sigma_{(t)} | 0 \leq t \leq t_1)) \quad (14)$$

The time-dependent fracture likelihood of each layer was formulated with Equation (14). Weibull reliability function implies that if a specimen survives after being tested with a particular tensile stress, it never fails upon experiencing any stress less than that testing stress. This definition departed from the uncertainty on the presence of the weakest link, i.e., pre-existing macroscopic flaws in SiC clad. These flaws were assumed as time-independent since radiation damage is not likely to contribute to the new formation of macroscopic voids in the clad. For that reason, the time-dependent failure probability was based upon the

maximum stress the clad experiences during a period. Such definition resulted in a monotonically increasing time-dependent clad fracture probability as suggested by Equation (14).

2.3.2. Modeling the loss of clad functions

After the fracture probability for each cladding layer was calculated, the failure probability for the whole multi-layered SiC cladding was estimated. This whole-clad failure has been defined with different assumptions in previous studies. Lee [6], [7] assumed that duplex and triplex clad fail when either the inner CVD monolith or the CVI matrix composite layer fractures. Under this failure definition, the triplex clad almost certainly fail during normal reactor operation, while the duplex clad failed when the fuel rod internal pressure rose slightly above its design criteria [6]. Conversely, Deng [9] assumed that clad fails when all of its layers fail, which resulted in $\sim 1\text{E-}9$ duplex clad failure in a steady-state reactor condition. These assumptions led to diverging conclusions: either the SiC clad is too fragile or extremely reliable as a nuclear fuel cladding. To resolve such difference in perspectives, the clad failure definition is revisited in this section based on material properties of each layer.

Characteristics of each SiC clad layer were described in previous studies [2], [5], [7], [25]. CVD SiC provides an excellent heat transfer capability, gas hermeticity, oxidation resistance, and a decent structural strength capability. The CVI SiC matrix composite complements these properties by improving the clad's strength at the cost of a reduced overall thermal conductivity. Although the composite matrix is hermetic at normal reactor operation condition, there is no data to support this behavior at accident conditions.

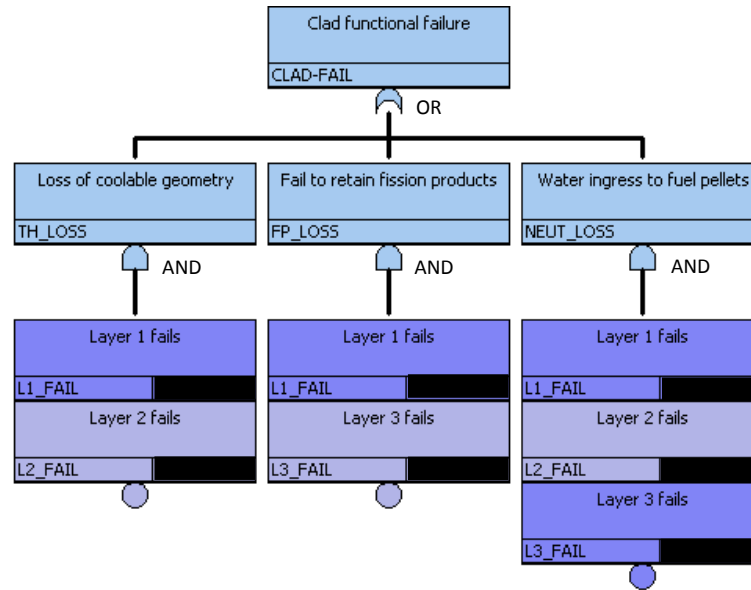
Based on the aforementioned material properties, if any of inner CVD or CVI matrix composite fails, the SiC clad gradually loses its load-bearing capability. Therefore, it can be recognized that the clad failure definition proposed by Lee [6] emphasized on the earliest onset of loss of coolable geometry. This definition may be too conservative in accident scenarios, because the reactor safe shutdown state may be achieved before the clad's degradation begin to alter the fuel rod's geometry. Additionally, the clad is not expected to degrade further in a safe shutdown condition as indicated by Equation (14). On the other hand, the failure definition proposed by Deng [9] provides a different insight. This failure event, i.e., the event where all layers fracture, marks the latest possible time when all of clad functions are lost, including the coolable geometry and fission gas retention functions. This event may be too optimistic, since failures on any of the functions may precede this event. Bearing these considerations, it is deemed necessary to refine the failure definition for multi-layered SiC clad in accident conditions.

The clad failure was redefined by recalling the main functions of a nuclear fuel cladding, i.e., (1) Maintaining coolable geometry to allow for decay heat removal, (2) Acting as a barrier to prevent fission products release to the coolant, and (3) Preventing coolant water ingress to fuel pellets, since it can change thermal neutron flux distribution. These functions for the triplex clad is tabulated in Table 2. The constituent layers for each function were assigned qualitatively based on the aforementioned discussion on layers' characteristics with additional assumptions. Layer 2 was conservatively assumed to be non-hermetic in accident scenarios. Layer 3 was reasonably assumed to possess no load-bearing capability due to its thin structure.

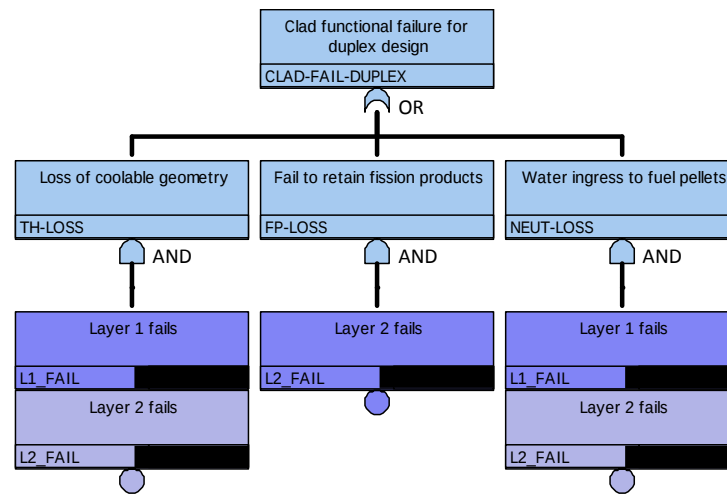
Table 2. Description of clad functions for triplex SiC cladding

	L1: CVD-SiC	L2: CVI SiC matrixcomposite	L3: CVD SiC
Function of individual layer	Fission gas retention and load sharing	Primary load-bearing structure and catastrophic shatter prevention	Separation of L2 from H ₂ O and fission gas retention
Overall clad function	Performed by		
Retain fission products	Yes	No	Yes

Maintain coolable geometry	Yes	Yes	No
H ₂ O-UO ₂ gap separator	Yes	Yes	Yes



(a) Triplex structure



(b) Duplex structure

Figure 6. Fault Tree representation of a multi-layered SiC clad functional failure

Fuel cladding was defined to be in an excellent working condition in this study if all of its functions listed in Table 2 are satisfied. With such a definition, fuel cladding functional failure occurs if any of its functions is lost. This logic was brought into a Fault Tree model shown in Figure 6. With a similar logic, the duplex clad functional failure lies entirely on the outermost CVD SiC layer (i.e., Layer 2). With this model, clad functional failure probability was formulated in the Boolean format as shown in Equation (15)

and in the arithmetic format as given in Equation (16). These equations were used to estimate clad functional failure at each i out-of n_z axial volumes. Since failure can occur at any axial location, the overall clad failure probability was calculated with Equation (17).

$$P_{fi_triplex} = [P_{f,1}(P_{f,2} + P_{f,3})]_i \quad (15)$$

$$P_{fi_duplex} = P_{f,2i}$$

$$P_{fi_triplex} = [P_{f,1}(P_{f,2} + P_{f,3} - P_{f,1}P_{f,2}P_{f,3})]_i \quad (16)$$

$$P\left(\bigcup_{i=1}^{n_z} P_{fi}\right) = 1 - P\left(\bigcap_{i=1}^{n_z} P_{fi}\right) = 1 - \prod_{i=1}^{n_z} (1 - P_{fi}) \quad (17)$$

2.4. Code Coupling

A fuel performance code to calculate fuel rod's temperature and stress distributions, as well as the resulting clad failure probability, was developed in MATLAB. This code imported the reactor parameters computed by RELAP5 [21]. RELAP5 facilitates a single-layered cladding structure in its fuel model. To ensure an agreement on the amount of heat transferred through the single and multi-layered clad structures, representative values of heat transfer coefficients were estimated for the RELAP5 fuel model. To calculate these coefficients, first consider the steady-state equation for the clad region:

$$-\frac{1}{r} \frac{d}{dr} \left(k_{eff} r \frac{dT}{dr} \right) = 0 \quad (18)$$

Which can be integrated into:

$$k_{eff} \Delta T_{clad} \equiv k_{eff} (T_{R4} - T_{R1}) = \frac{q'}{2\pi} \ln \left(\frac{R_4}{R_1} \right) \quad (19)$$

Heat conduction through the multi-layered clad is taken from reference [6]:

$$T(r_1) = \frac{q'}{2\pi} \left[\frac{1}{k_1} \ln \left(\frac{R_2}{r_1} \right) + \frac{1}{k_2} \ln \left(\frac{R_3}{R_2} \right) + \frac{1}{k_3} \ln \left(\frac{R_4}{R_3} \right) \right] + T(R_4), \quad R_1 \leq r_1 \leq R_2 \quad (20)$$

By substituting r_1 in Equation (20) with R_1 , the linear heat rate through the multi-layered clad was equated with the one in the single-layered clad in Equation (19) as:

$$q' = \frac{2\pi k_{eff} \Delta T_{clad}}{\ln \frac{R_4}{R_1}} = \frac{2\pi \Delta T_{clad}}{\sum_{i=1}^3 \frac{\ln \left(\frac{R_{i+1}}{R_i} \right)}{k_i}} \quad (21)$$

Such that effective thermal conductivity can be found with Equation (22). Meanwhile, effective volumetric heat capacity can be derived from the definition of heat capacity, i.e., the amount of energy required to raise the clad temperature by 1 °K. The energy to raise the single-layered clad's temperature should equate the energy required to heat up each individual clad layer to the same temperature as given in Equation (23) to (24). Therefore, the effective volumetric heat capacity was defined with Equation (25).

$$k_{eff} = \frac{\ln \frac{R_4}{R_1}}{\sum_{i=1}^3 \frac{\ln \left(\frac{R_{i+1}}{R_i} \right)}{k_i}} \quad (22)$$

$$Q_{eff} = Q_1 + Q_2 + Q_3 \quad (23)$$

$$(\rho \times C_p)_{eff} V_{clad} = \sum_{i=1}^3 (\rho \times C_p)_i \times V_i \quad (24)$$

$$(\rho \times C_p)_{eff} = \frac{\sum_{i=1}^3 (\rho \times C_p)_i \times (R_{i+1}^2 - R_i^2)}{R_4^2 - R_1^2} \quad (25)$$

RELAP5 model used in this study was a two-loop 1000 MWe PWR model [26]. The fuel performance code imported RELAP5 output parameters, i.e., linear heat generation rate q' and reactor power history $P(t)$ as the heat source boundary conditions, and heat sink boundary conditions, i.e. coolant pressure, coolant liquid temperature, void fraction, gas/steam temperature, and the heat transfer coefficient at the clad-coolant boundary.

2.5. Selection of SiC Parameters

2.5.1. Thermal conductivity

CVD SiC thermal resistivity as a function of irradiation temperature was reported in a reference [27]. However, it had significant scatter deviations and was measured from plate specimens which may not be representative for tube geometries. Therefore, a constant thermal conductivity of 9.5 W/mK was selected for CVD SiC in this study following reference [6], [7], [23]. This value is slightly more conservative than the 10 W/mK conductivity used in [9]. Meanwhile, the thermal conductivity of SiC composite was updated to 4.5 W/mK following reference [28]. This value has a better statistical confidence from a previous 2011 report [23].

2.5.2. Heat capacity and density

Temperature-dependent heat capacity for both CVD and CVI matrix composite [28] is given in Equation (26). The density of CVD SiC was reported as 3170 kg/m³ and 3210 kg/m³ in [6] and [24] respectively. The first value was arbitrarily selected since there is no significant difference between the two. Meanwhile, the composite density of 2650 kg/m³ was adopted from reference [28] which founded the basis for calculating the composite thermal conductivity.

$$Cp \left[\frac{J}{kg \cdot K} \right] = 925.65 + 0.3772T - 7.9259 \times 10^{-5}T^2 - 3.1946 \times 10^{-7}T^{-2} \quad (26)$$

2.5.3. The coefficient of thermal expansion (CTE)

CTE of CVD SiC was adopted from correlation developed by Snead [29] shown in Equation (27). Meanwhile, CTE of CVI matrix composite was updated from a recent report [27] as given in Equation (28). This temperature-dependent CTE is more relevant for accident scenarios compared to the static value used in the previous study [7].

$$\alpha \left[\frac{10^{-6}}{K} \right] = -1.8276 + 0.0178 \times T - 1.5544 \times 10^{-5} \times T^2 + 4.5246 \times 10^{-9} \times T^3, \quad (27)$$

$$125 \text{ K} \leq T \leq 1273 \text{ K}$$

$$\alpha = 5 \times 10^{-6}, \quad T > 1273 \text{ K}$$

$$\alpha \left[\frac{10^{-6}}{K} \right] = -0.7765 + 1.435 \times 10^{-2} \times T - 1.2209 \times 10^{-5} \times T^2 + 3.8289 \times 10^{-9} \times T^3 \quad (28)$$

2.5.4. Young's modulus and Poisson ratio

The elastic modulus for CVD SiC and composite in this study were taken as 460 GPa and 230 GPa respectively and were assumed as isotropic [6], [9], [24], [29]. The CVD SiC and composite Poisson ratio were adopted from the same set of references as 0.21 and 0.13 respectively. Mieloszyk [24] explained that the variation of these parameters due to irradiation and temperature is trivial. Therefore they can be assumed as constants.

2.5.5. Volumetric swelling due to irradiation

This study adopts the most recent volumetric swelling model as a function of the neutron dose and irradiation temperature [27], which is applicable to CVD SiC, HNS fiber reinforced CVI SiC/SiC composite plates and tube specimens, and CVI SiC/SiC composites with SA3 fibers. The correlation was used to build a swelling response surface shown in Figure 7. A dose-saturated swelling correlation was fitted from this response surface as given in Equation (29), where T indicates the steady-state clad temperature.

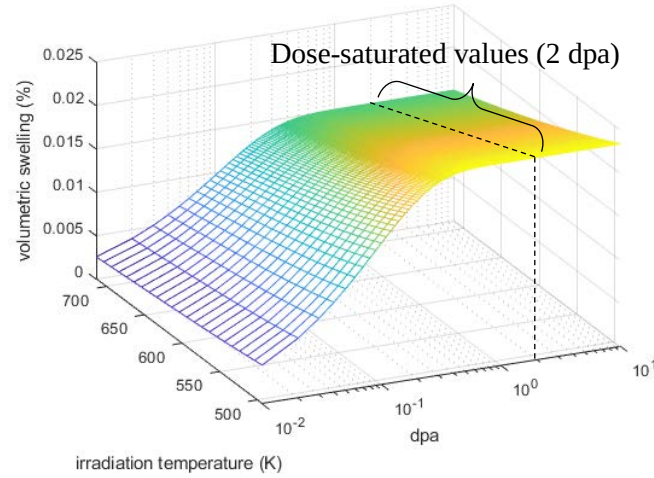


Figure 7. Volumetric swelling response surface highlighting the dose-saturated swelling regime

$$S \left[\frac{\Delta V}{V} \times 100\% \right] = 3.653 \times 10^{-8} T^2 - 8.123 \times 10^{-5} T + 0.05447 \quad (29)$$

Comparison of clad swelling in a reference EOL scenario [6] calculated using Lee's model [6] and Equation (29) is shown in Figure 8. Both models agreed that irradiation swelling contradicts the temperature distribution. However, swelling calculated with Equation (29) has continuous transitions at layer boundaries, since the model applies to both CVD and CVI composite.

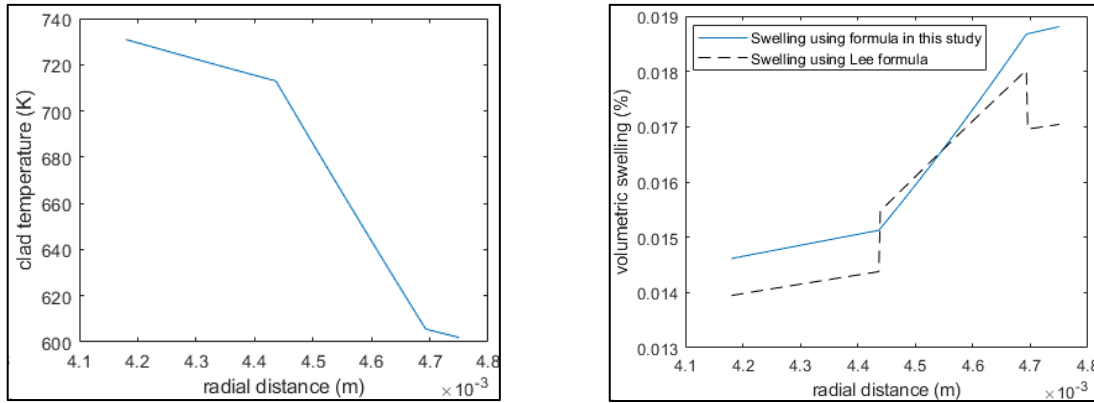


Figure 8. Radial temperature distribution (left) and clad irradiation swelling (right)

2.5.6. Weibull parameters

Table 3 summarizes the parameters for CVD SiC and CVI matrix composite separately. Reference [27] provides Weibull parameters for duplex and triplex SiC clad as a whole. However, these parameters are valid if the stress is applied uniformly throughout the specimen. This stress uniformity is unlikely in reactor operations and therefore the integrated Weibull parameters for multi-layered cladding is not considered in this study.

Table 3. Literature survey of Weibull parameters

Reference	Weibull slope parameter m	Weibull scale parameter σ_0	Specimen dimension
Lee [6]: CVD SiC	7.5	369 MPa	Volume: 6.36E-10 m ³
Katoh [23]: CVD SiC irradiated at 300 °C	2.7	506 MPa	50.2 mm x 6.35 mm x 2.63 mm

Katoh [23]: CVD SiC irradiated at 650 °C	3.2	429 MPa	
Deng [30]: CVD SiC	3.82	172.93 MPa	Area: 126.16E-6 m ²
Lee [6]: CVI SiC composite	17.5	290 MPa	15 mm x 3 mm x 2.33 mm
Singh [31]: CVI SiC composite	10.13	249 MPa	Outer diameter 9.53 mm Inner diameter 7.97 mm Length 150 mm

A failure probability plot using the collected Weibull parameters was generated in Figure 9. There is a general agreement on the failure likelihood for CVD using Lee's parameters [6] and Katoh [23], and for CVI matrix composite using Lee's [6] and Singh's [31] parameters. Deng's parameters [30] implied that CVD SiC has a better fracture toughness than the CVI matrix composite. Such condition seems unlikely and therefore we focused our attention to Katoh [23] and Lee [6] CVD parameters. Lee [6] adopted results from an experiment [29] predating Katoh's [23]. For that reason, Katoh parameters was selected albeit with a slight adaptation to adjust it to the average clad temperature in this study. The averaged CVD parameters were $m=2.92$ and characteristic strength=467 MPa. Meanwhile, CVI parameters were adopted from Singh [31] since they are the most recent and are considered "one of the best data for the modeling purposes currently due to limited studies on statistics of mechanical properties of SiC/SiC tube" [27].

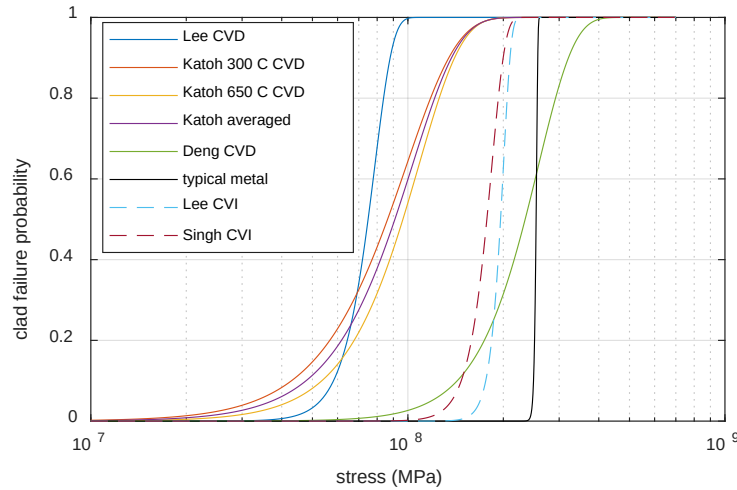


Figure 9. Comparison of failure probabilities from various sets of Weibull parameters

3. Results and Discussions

3.1. Reference LBLOCA Scenario Case Study

A cold-leg LBLOCA was simulated with a double-ended guillotine break at the cold leg of primary Loop 1. Active ECCS (high-pressure and low-pressure active injections) was activated 30 seconds after LBLOCA was detected. This time range is the typical delay time to start emergency diesel generators and pumps in the current PWR with UO₂-Zr system [32]. The active fuel assembly was modeled into 12 axial nodes and 8 radial nodes in RELAP5. The effective radial thermal conductivity for the triplex SiC cladding was calculated as 6.387 W/mK.

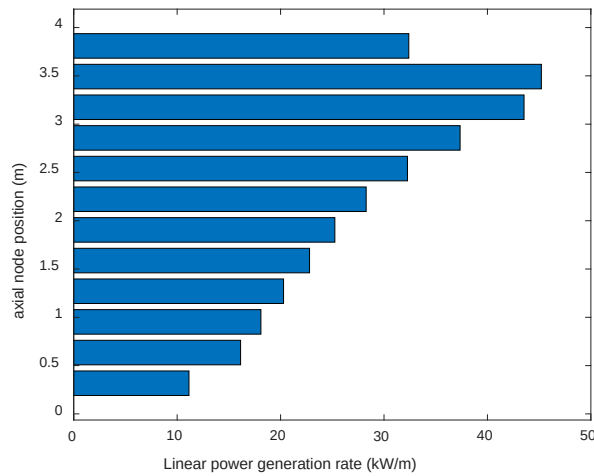


Figure 10. Axial power distribution in the hottest fuel rod

The axial power peaking distribution for the hottest $\text{UO}_2\text{-Zr}$ fuel rod is shown in Figure 10, which indicates that the hottest axial location is located near the top of fuel assembly. The axial power of $\text{UO}_2\text{-SiC}$ fuel rod was assumed to follow this distribution as well. Despite this axial power variation, heat conduction and changes of thermal deformation along the principal axial axis were not modeled. It can be reasonably assumed that the temperature distribution along the axial axis has negligible contributions to the clad fracture mechanics in this LOCA accident scenario.

Consider a comparison between radial and axial thermal gradients at the hottest location in the cladding which nodes are illustrated in Figure 11. The mid-top inner clad section was assumed as the axial heat source due to its axial power peak, such that extreme thermal gradients may occur over the cladding length between this node to the bottom axial node which is cooled by ECCS, or in a more localized manner between this node and any other axial nodes. Maximum values of these local axial gradients were selected at each time step throughout the accident progression. Likewise, the radial thermal gradient was measured throughout the clad thickness, and locally within the inner monolith only. The selection of maximum local axial gradients and constant location of local radial thermal gradient aggressively challenged our assumption, because the low localized radial thermal gradient due to the high CVD's thermal conductivity was compared against the maximum values of local gradients from various axial node pairs.

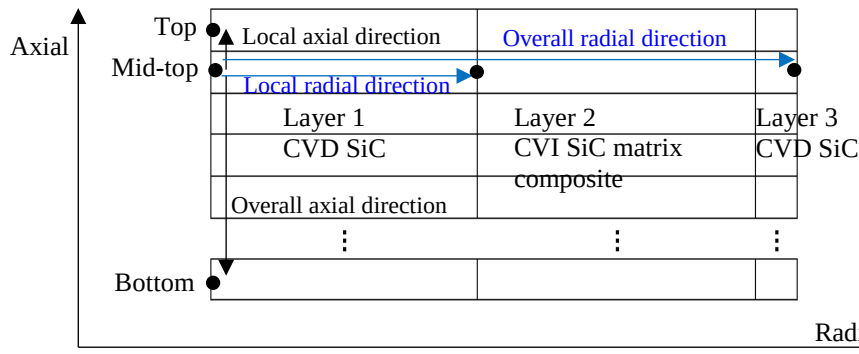


Figure 11. Locations of nodal test points and directions of heat transfer in the axial and radial principal axis

The four aforementioned thermal gradient measurements were contested during the reference LBLOCA scenario as shown in Figure 12. Thermal gradient in the global cladding dimension was dominant in the radial direction, with a magnitude of ~ 2 to 4 orders higher than the axial direction. This observation

suggests that the overall thermal deformation gradient in the axial direction is trivial. However, there may also be a possibility that thermal gradients in localized observation points deviate from this expectation. It was observed that the local radial heat gradient was only moderately higher than its axial counterpart, which may raise doubts on our assumption. Despite this skepticism, the assumption is still generally valid since thermal stresses did not dictate the total stress as shown in Figure 13. Instead, radiation swelling and pressure loading governed the total stress and cladding tensile state in the inner clad layer and compressive state at the outer one throughout the accident. Therefore, it was reasonable to believe that unless there is an extreme temperature variation along nearby axial nodes in the cladding, SiC clad reliability can be sufficiently estimated with a 1-dimensional radial model of temperature and thermomechanical stress calculations.

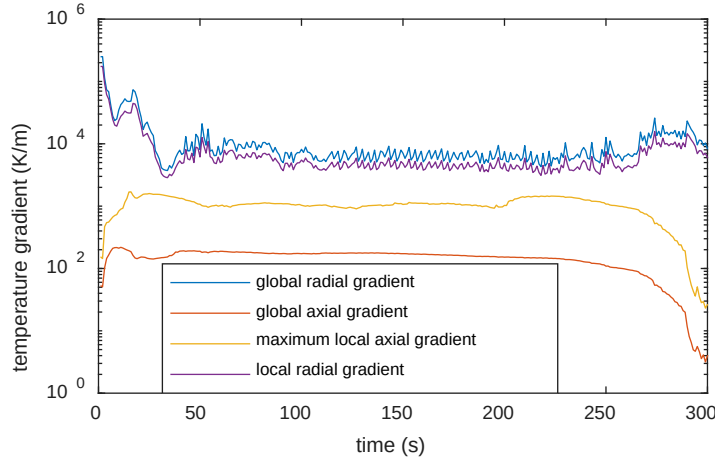
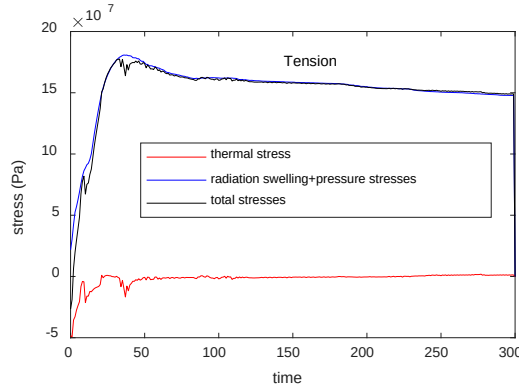
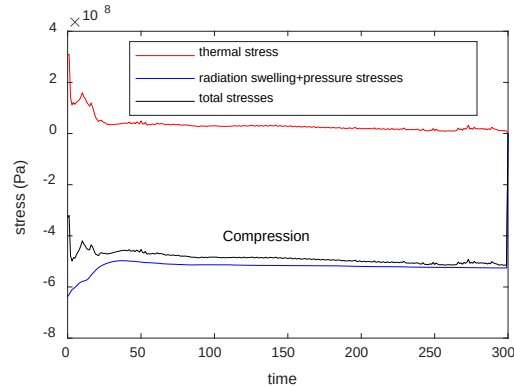


Figure 12. Cladding thermal gradient in the reference LBLOCA scenario



(a) Stresses on inner cladding surface



(b) Stresses on outer cladding surface

Figure 13. Comparison of thermal stress to the total stress at the inner and outer cladding surface during the reference LBLOCA scenario

3.2.1. Reactor Parameters

Reactor parameters during the reference LBLOCA accident scenario are shown in Figure 14 and Figure 15. Active ECCS (high pressure and low-pressure active injections) were activated 30 seconds following the pressurizer's low pressure signal. This injection delay is made possible by the use of passive accumulators [32]. As seen in the figure, passive accumulator tanks rapidly injected a large amount of borated water for ~1 minute. The integral of this passive injection curve equaled to ~106 m³ of borated water, which was twice the capacity of an accumulator tank in our reference plant. This amount of water was provided by two accumulator tanks at primary Loop 2. No credit was given to the broken Loop 1 to provide

cooling water to the core. Reactor thermal power was initially 2815 MWth, which rapidly decreased to 160 kWth decay heat at t=37 seconds. This amount of remaining decay heat was removed by active ECCS. Both low and high-pressure active ECCS systems were initiated, although only the low-pressure active ECCS system was credited in the plant Probabilistic Risk Assessment (PRA) model to successfully cool down the core due to its high mass flow rate.

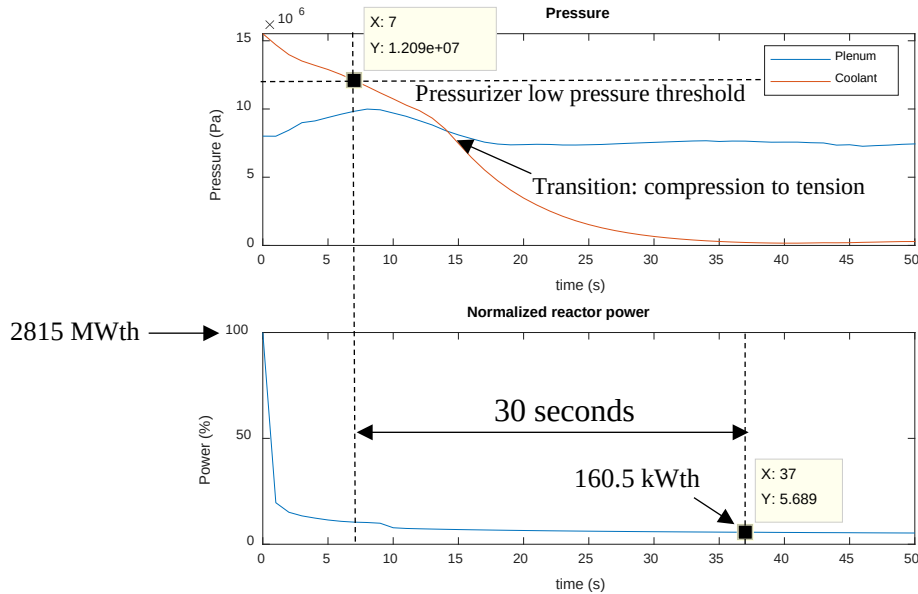


Figure 14. Reactor power and pressure during reference LBLOCA scenario

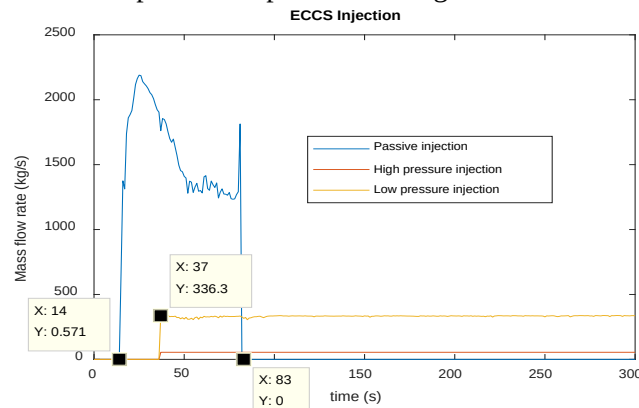


Figure 15. ECCS injection in the reference LBLOCA scenario

3.2.2. Fuel Parameters

Fuel parameters at the steady-state condition just before LBLOCA occurs are shown in Figure 16. Temperature distribution shows the hottest axial location corresponding to the axial peaking factor shown in Figure 10. Layer 1 was exposed to the highest temperature and therefore experienced more thermal expansion compared to other layers. However, it was constrained by Layer 2 which did not expand as much due to its lower temperature. As a result, Layer 1 experienced compressive (negative) thermal stresses in the hoop and axial direction. In contrary, Layer 3 experienced tensile (positive) thermal stresses in the same principal directions due to being forced to expand by Layer 2. This stress pattern was reversed for irradiation swelling stresses because the swelling is inversely proportional to irradiation temperature. For that reason, swelling and thermal stresses tend to neutralize each other. However as seen from the figure, swelling stresses slightly overpowered thermal stresses, shifting the total stress in the swelling's favor. The third stress component due to pressure loading was less than the thermal and swelling stresses. Pressure stresses during

steady state were compressive because the PWR coolant pressure was higher than the fuel plenum pressure. The resultant hoop and axial stresses put Layer 1 in tension and the other two layers in compression. Meanwhile, the high temperature gradient along the radial direction and compressive pressure stress caused all layers to experience compressive total radial stress. This observation was in agreement with results in a previous study [6].

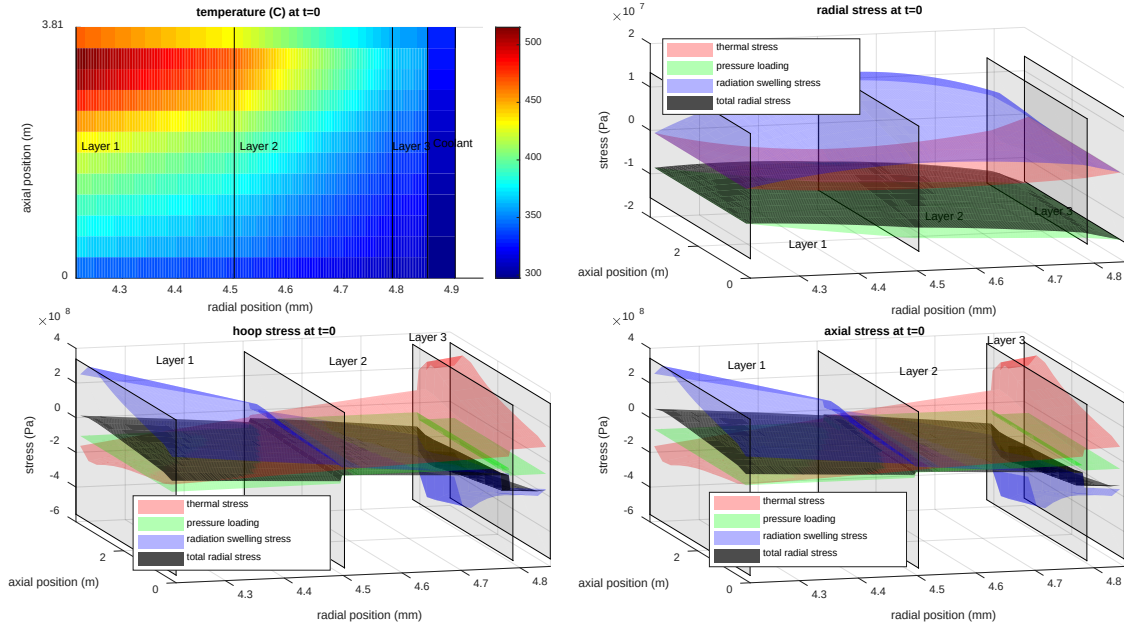


Figure 16. Triplex fuel parameters at beginning of LBLOCA (t=0): temperature distribution (top left), radial stresses (top right), hoop stresses (bottom left) and axial stresses (bottom right)

In the early seconds of LBLOCA accidents, the absence of fission power and the flash boiling of saturated core inventory caused the radial temperature gradient to be almost uniform, rapidly neutralizing the thermal stresses. As thermal stresses decreased and swelling remained constant in the absence of irradiation, the total stress eventually converged to the swelling stresses. This stress convergence is shown in Figure 17. Swelling stress continued to dictate the total stress until a significant portion of the core was significantly uncovered, after which the fuel internal pressure gradually rose and added tension to the clad.

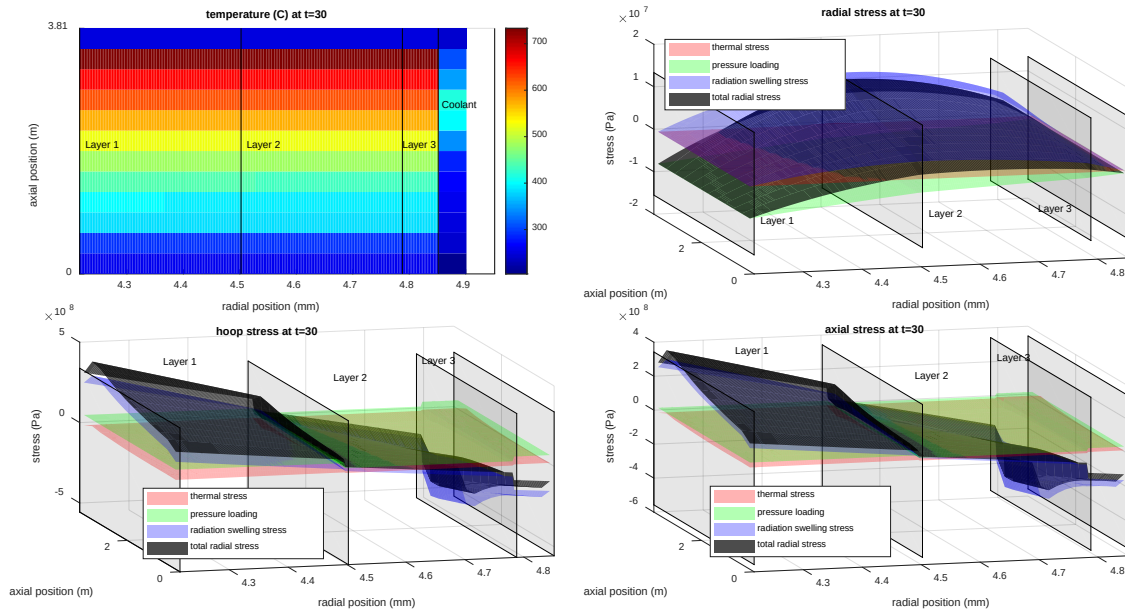


Figure 17. Triplex SiC clad parameters at t=30 seconds

Maximum temperature in the fuel rod during the LBLOCA accident scenario are shown in Figure 18. This figure suggests that neither fuel pellet nor SiC clad failed deterministically through thermal degradation. Meanwhile, time-dependent triplex clad fracture probabilities are shown in Figure 19. Layer 1 was more likely to fail compared to the other layers because it experienced tensile swelling stress. As a consequence, the clad failure probability calculated using Lee's conservative assumption [6] was ~ 1 (0.9997). It suggests that triplex SiC clad is not a promising ATF candidate. The clad functional failure probability calculated with the proposed model and Deng's model were $\sim 1\text{E-}6$ and $\sim 1\text{E-}15$ respectively. Both of these estimates were less than the initial conditional core damage probability for Zr cladding, which is in the order of $1\text{E-}2$ to $1\text{E-}3$. Therefore, the triplex SiC cladding provided a considerable safety benefit under these failure models. Future research is recommended to verify the suitability of these clad functional failure models with experimental findings.

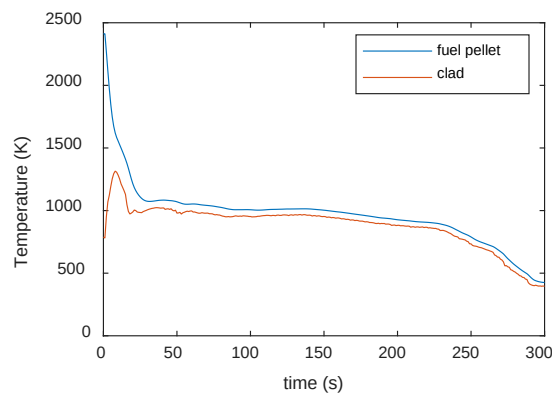


Figure 18. Maximum temperatures in the hottest fuel rod during LBLOCA scenario

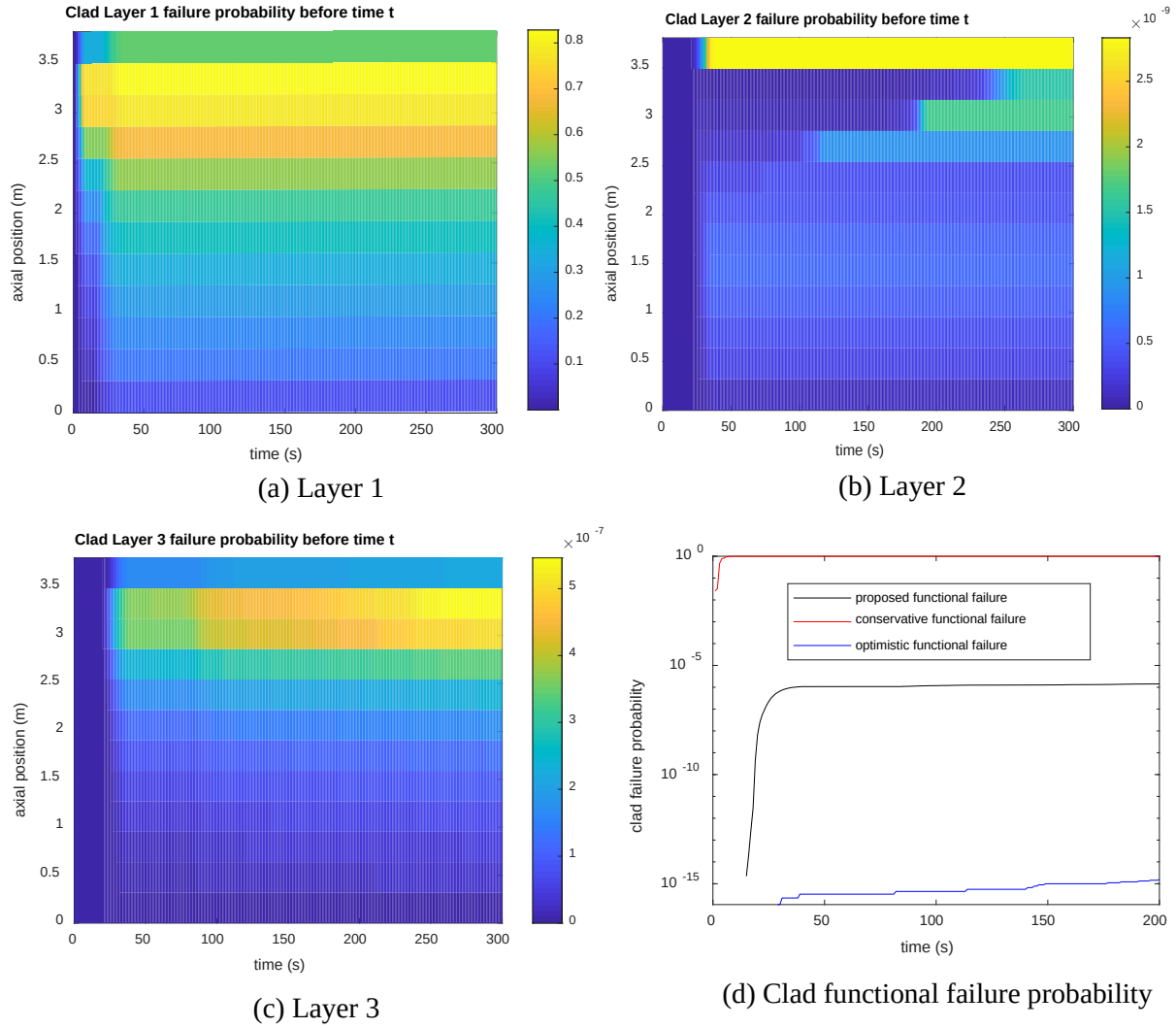


Figure 19. Time-dependent failure probabilities of triplex clad

As shown in Figure 19, Layer 1 (CVD SiC in triplex design) was prone to fractures. Furthermore, the most vulnerable axial position was found at the highest axial power peaking location. This vulnerability was caused by the tension dictated by the swelling stress profile as discussed above. To remedy this issue, Lee [6] suggested a duplex cladding design using a CVI SiC matrix composite as the inner layer and a CVD SiC outer layer with a 50:50 radius configuration. This design leverages the superiority of CVI matrix composite's load-bearing capability to handle the tensile swelling stresses. Unfortunately, it reduces the clad's heat transfer performance. The duplex's thermal conductivity was 6.0317 W/mK, compared to triplex SiC of 6.387 W/mK. Duplex SiC also has a lower effective heat capacity compared to triplex design as shown in Figure 20. The temperature and stress distribution on the duplex cladding is shown in Figure 21. As expected, the temperature gradient across radial direction was higher in duplex cladding compared to the triplex, due to its lower effective thermal conductivity. Layer 1 experienced net tensile stress in the axial and hoop principal directions as anticipated.

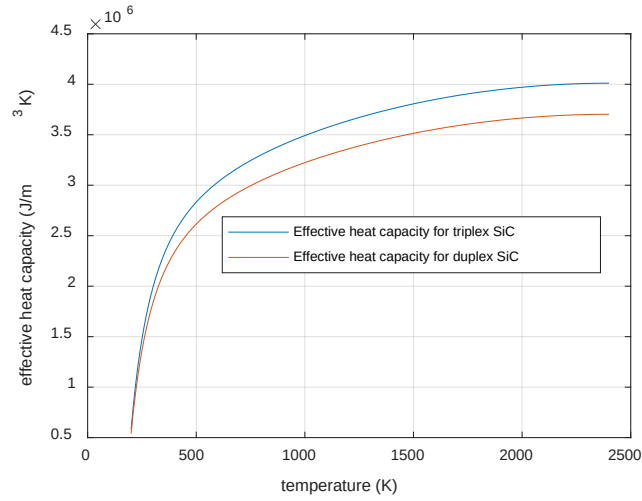


Figure 20. Comparison of effective heat capacity for triplex and duplex SiC

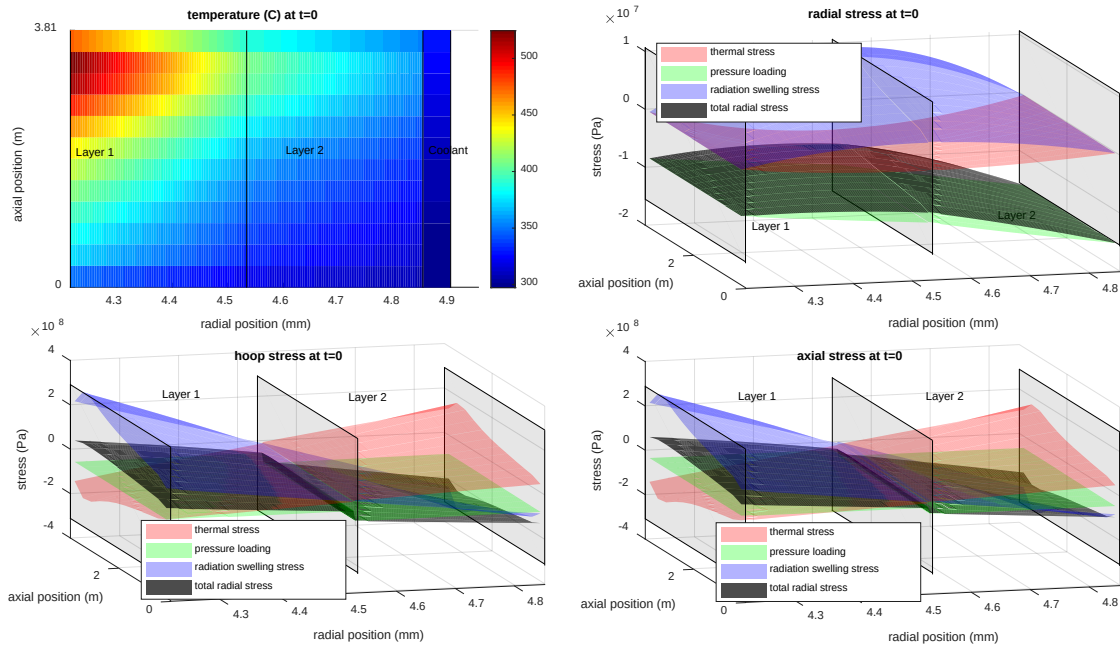


Figure 21. Duplex SiC clad parameters at beginning of LBLOCA accident ($t=0$): temperature distribution (top left), radial stresses (top right), hoop stresses (bottom left) and axial stresses (bottom right)

Failure probability of each layer and the overall duplex cladding function is shown in Figure 22. As expected, the reliability of Layer 1 in duplex clad was better than the Layer 1 of triplex SiC, albeit slightly. The clad functional failure calculated with the Lee's model [6], proposed and Deng's [9] models were ~ 1 (0.97), $\sim 3\text{E-}2$ and $\sim 1\text{E-}2$ respectively. With Lee's model, duplex SiC clad was slightly better than the triplex counterpart although it did not change the implication that clad failure was almost certain. Meanwhile, under the other two perspectives on clad failure, duplex SiC design was less reliable than the triplex SiC. This reliability loss might be attributed to the absence of redundancies in fission product retention as shown in Figure 6.

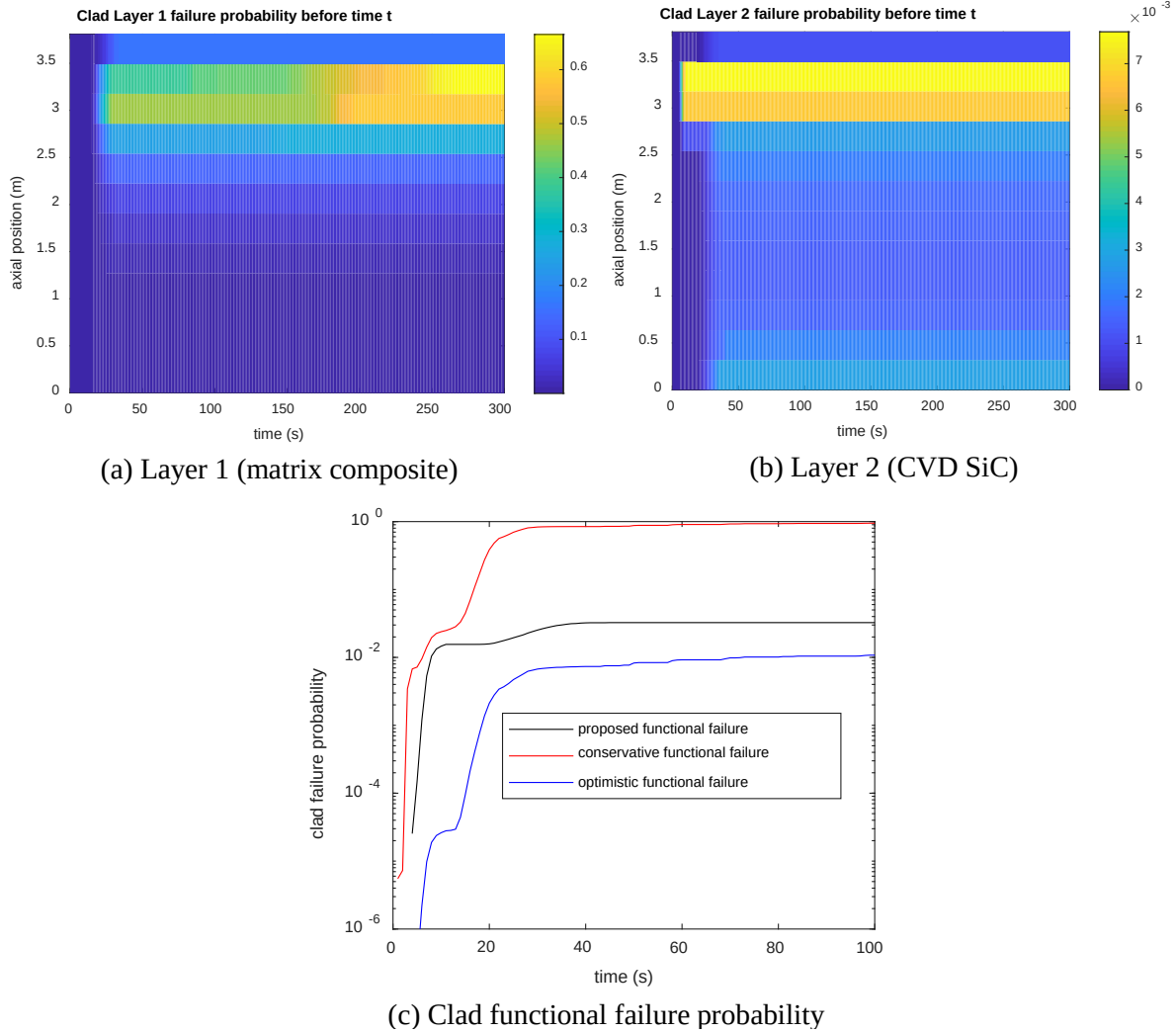


Figure 22. Time-dependent failure probabilities of duplex SiC cladding

3.2.3. Discussion: Recovery of volumetric irradiation swelling

Results and aforementioned discussions suggest that multi-layered SiC clad reliability is driven mainly by irradiation swelling. It was noted that this swelling may be recovered or annealed with thermal exposure. During steady state operation, the presence of irradiation continuously forms point defects which create interstitials and vacancies in the crystalline lattice. These defects expand the cladding resulting in a clad volumetric swelling. As irradiation stops during accident mitigation, remaining decay heat may provide energy for the C and Si Frenkel pairs to overcome the activation energy and recombine. As a result, the swollen clad may shrink.

Idris [33] [34] reported the swelling recovery rate and activation energies for CVD SiC and CVI SiC composites. This swelling recovery may alter the reliability estimation of SiC clad. However, swelling recovery is a relatively slow process compared to fast transients such as LBLOCA. The swelling recovery rate reported in Idris studies was in the order of several hours under high temperature (above 1000 K) [33] [34]. Even if ECCS is delayed in the order of hours to allow swelling recovery to take place and decrease clad fracture likelihood, the fuel decay heat would have melted the fuel pellets causing a deterministic failure instead. For example, consider an extreme case where active ECCS is not activated until 20 minutes after LBLOCA. Swelling recovery within 20 minutes for inner monolith as the most vulnerable layer can be roughly approximated using the CVD swelling recovery rates at 1073 K in Idris study [33]. In 20 minutes,

the CVD layer would have shrunk by 0.6 % of its irradiated-swollen volume. The 1-dimensional swelling in each principal direction is a third of this volumetric swelling, which renders it more insignificant. A separate simulation in RELAP5 confirmed that fuel pellet indeed reached its melting temperature in this extreme LBLOCA scenario. This example shows that swelling recovery plays a trivial role in our case studies and therefore was omitted to simplify the fuel model. Swelling recovery may be an essential consideration in slower transients such as Small-Break LOCA or steam generator tube rupture. This hypothesis is left for future research. The constant swelling assumption in this study is deemed conservative since it maintains tensile stresses which makes the inner monolith (Layer 1) vulnerable. This assumption is in agreement with a previous study [24].

3.3. Extended LBLOCA Case Study

Results in the previous section suggest that SiC cladding is reliable enough in the reference LBLOCA scenario. With such consideration, active ECCS actuation timing may be reconfigured by allowing longer deviations without increasing plant risk, i.e., achieving the design criteria in Equation (2). For that reason, the fuel's response was analyzed in various ECCS activation scenarios. The Peak Cladding Temperature (PCT) of Zircaloy and SiC cladding are compared in Figure 23. This figure confirms our hypothesis as illustrated in Figure 3, that SiC cladding increases the time margin before fuel fails through thermal degradation as compared to Zr cladding. This time extension was approximately 15 minutes. If PCT is used as the sole failure criteria, ECCS should be activated within a deterministic time limit of 20 minutes after LBLOCA. However, SiC clad may statistically fracture before reaching its decomposition temperature, which implies that ECCS activation criteria may instead be probabilistic to limit the likelihood of such brittle fracture events.

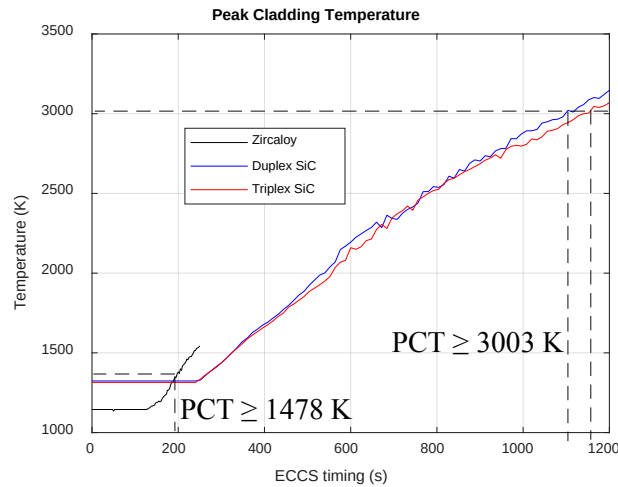


Figure 23. Peak Cladding Temperature (PCT) of clad designs

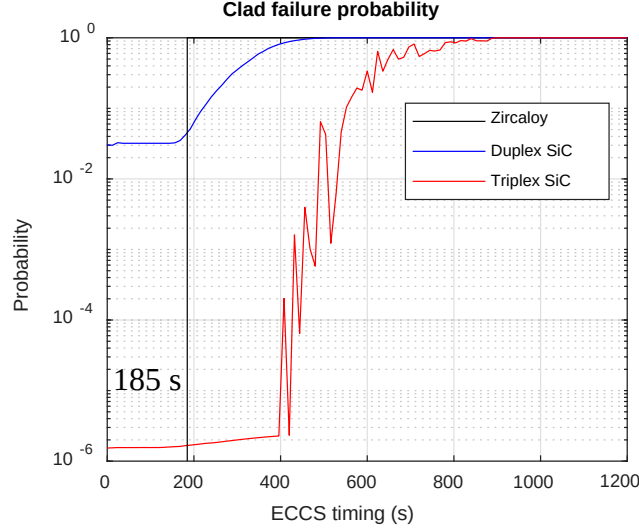
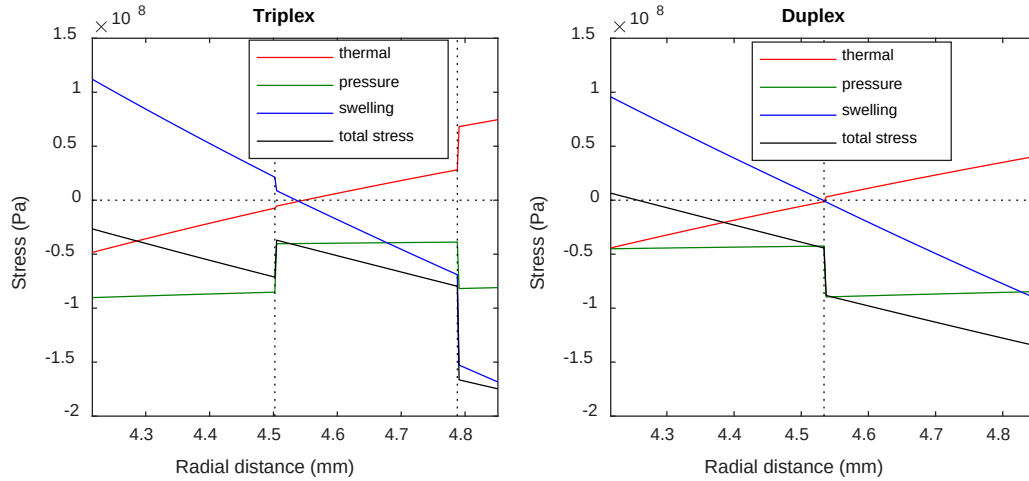


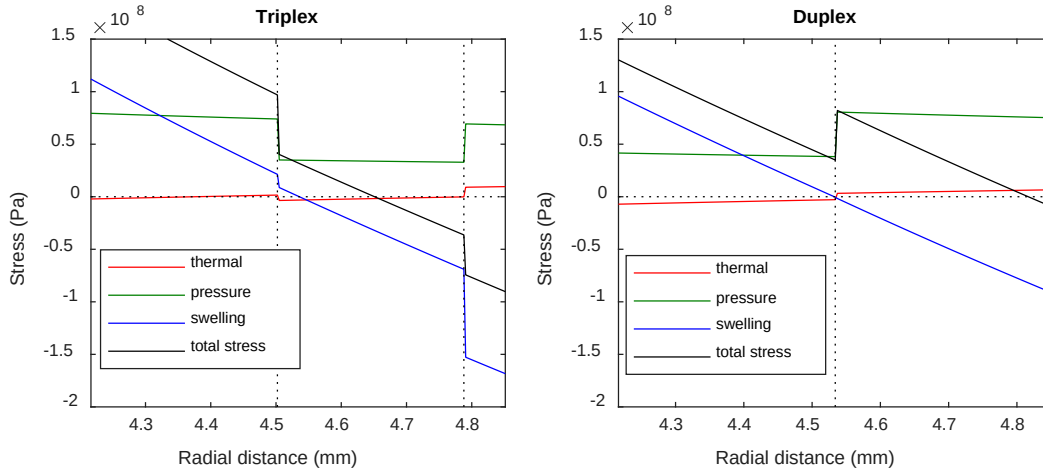
Figure 24. Clad functional failure probability

Clad functional failure probability due to brittle fractures of each layer is shown in Figure 24. By comparing this figure with Figure 23, it can be concluded that the clad failure by brittle fracture preceded the failure by thermal decomposition. Furthermore, the triplex clad design was more reliable than its duplex counterpart. Another notable observation from Figure 24 is that the failure probability of triplex SiC varied rapidly with actuation timing as opposed to the gradual failure profile of the duplex structure. To explain this result, consider the stress components at the bottom section of the clad, i.e. the section that experienced a rapid quenching during late reflood events, as shown in Figure 25. This figure highlights hoop stresses only, as they sufficiently represent axial and radial stresses as suggested by Figure 16 and Figure 21. The main difference between the two clad designs is that the swelling stress compressed the outermost triplex layer more than it did to the duplex structure as seen in Figure 25(a). As the clad temperature decreased along the outward radial direction, the outermost layer swelled more than the inner one. As a result, the outer layer experienced a compressive force exerted by the inner layer. Logically, the outermost layer in the triplex structure received an increased compression because it was thinner than other layers, as opposed to the duplex clad. Analytically, the thin outermost triplex layer caused a reduction of the integral term in the swelling stress calculation [6] shown in Equation (30) compared to the duplex structure, which explains the increased compression.

$$\sigma(r)_{\theta,i,Th} = E_i \left[\left(\frac{1}{(1-v_i)r^2} \int_{R_i}^{R_{i+1}} \frac{S(T)}{3} r dr \right) - \frac{S(T)}{3(1-v_i)} + \frac{1}{1+v_i} \left(\frac{C_{1,s}^j}{1-2v_i} + \frac{C_{2,s}^j}{r^2} + \frac{v_i \varepsilon_{0,s}}{1-2v_i} \right) \right] \quad (30)$$



(a) Stresses at beginning of accident



(b) Stresses at t=300 seconds

Figure 25. Stress components at the bottom part of cladding before ECCS activation

As the swelling stress was constant throughout the accident scenario, it provided a better margin of fracture toughness to the outermost triplex layer than to the outermost duplex one. For example, consider the clad stresses at 5 minutes after LBLOCA detection, during which duplex clad reliability had degraded while triplex clad's had yet to as suggested by Figure 24. As shown in Figure 25(b), the thermal stresses were trivial which might be caused by the negligible radial temperature gradient. As the internal fuel gap pressure exceeded the coolant pressure as shown in Figure 14, it exerted tension on the cladding. In this condition, Layer 2 and Layer 3 of the triplex clad were sufficiently compressed while most of the duplex clad was under tension. Therefore, the triplex clad was still likely to maintain its coolable geometry function through Layer 2, and its fission gas retention function through Layer 3. Conversely, the duplex clad's fission gas retention capability started to degrade due to probable fractures in its CVD layer. For that reason, triplex clad was more reliable than duplex clad. When ECCS actuated after core uncover, the temperature within triplex's Layer 3 dropped which was not reciprocated by triplex's Layer 2 due to their difference in thermal conductivities. As a result, triplex's Layer 3 experienced tensile thermal stress which may offset the swelling strength margin and cause it to fracture. The extent of this thermal stress was uncertain due to the epistemic uncertainty in the boiling heat transfer model of RELAP5 [35]. For that reason, the failure likelihood of triplex clad's fission product retention fluctuated when ECCS actuation was delayed as seen in Figure 24. Meanwhile, duplex clad's fission gas retention capability might have been compromised by

the internal pressure buildup even before quenching events occurred, which explains the gradual duplex's failure profile in Figure 24.

The ECCS timing criteria for SiC clad was optimized with Equation (3). The reference ECCS distribution was reasonably assumed as Lognormal (3.2, 0.66), having a mean of 30 seconds and a $1E-3$ failure probability of timely activation, i.e., activation within 185 seconds after LBLOCA. It was found that ECCS could not be optimized to satisfy Equation (2) for the duplex clad. This result was because duplex clad failure increased monotonically starting from $1E-2$ as seen in Figure 24. Meanwhile, several possible ECCS activation densities for the triplex clad which gave a conditional core damage probability of $1E-3$ are shown in Figure 26. To reduce the plant risk further, one can lower the mean or deviation of the distribution parameters shown in the figure.

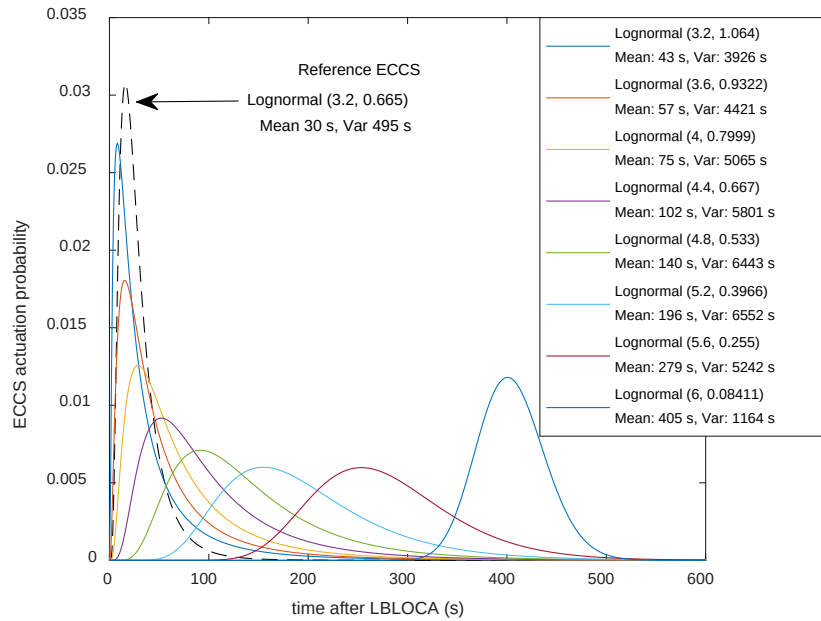
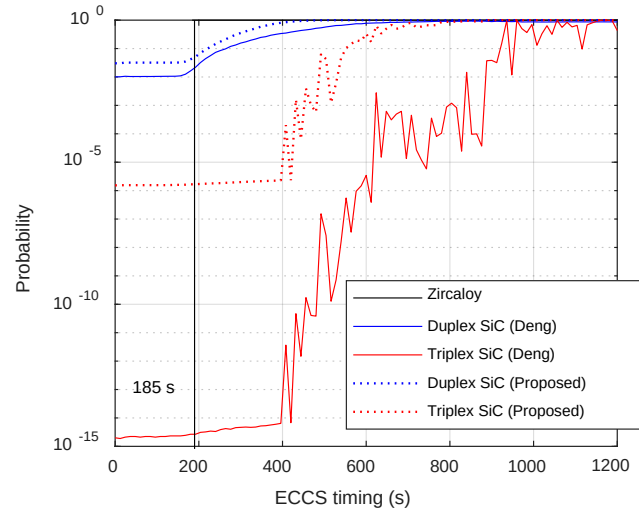
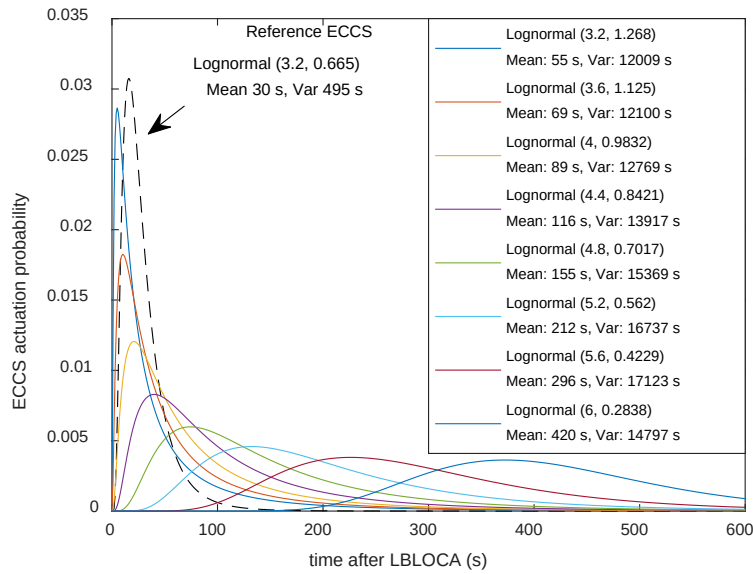


Figure 26. ECCS actuation density functions for triplex SiC clad

As a comparison, the clad functional failure probability as calculated using the optimistic failure definition (Deng [30]) is shown in Figure 27(a). Clad failure probability using the conservative failure assumption (Lee [6]) is not shown because the value approached unity regardless of the ECCS timing. As seen from the figure, the different clad failure assumptions led to a significant change in reliability estimates of the triplex clad only. The ECCS actuation timing densities which correspond to this triplex clad reliability profile are shown in Figure 27(b). As expected, the optimistic failure assumption provided further margins to relax the ECCS activation timing. However, an optimal ECCS configuration that reduces duplex clad damage probability less than $1E-3$ was not found.



(a) Clad functional failure probability



(b) ECCS actuation density function for triplex clad

Figure 27. Clad reliability calculated using the optimistic failure definition (Deng [30]) and the corresponding ECCS actuation density

Results obtained in this study indicated that triplex SiC cladding is more reliable than the duplex and Zr clads in LBLOCA scenarios. It was confirmed that the safety benefit of this triplex SiC cladding could be capitalized to relax the ECCS activation timing without increasing the plant risk. This relaxation allowed the ECCS mean activation time to be extended in the order of tens of seconds to ~7 minutes after LBLOCA. This result is comparable with the range of time extensions offered by FeCrAl and Cr-coated Zr ATFs in similar LBLOCA scenarios [15].

4. Conclusion and future works

This research investigated the failure of duplex and triplex SiC cladding in an LBLOCA accident in a PWR reactor. The results were used to optimize the ECCS performance criteria in terms of its actuation timing, such that the risk of the new plant design is less or equal to the risk of the initial plant configuration.

Several important conclusions were drawn based on our assumptions and hypothesis in this study. Results showed that clad failure by brittle fractures prevailed failure by thermal degradations. However, both triplex and duplex SiC clad reliability estimate varied significantly when different assumptions of clad failure were used. This failure definition is deemed a gray area having potentially significant impacts which makes it worthy of revisiting in future research. Under the proposed definition of clad failure, triplex SiC clad design was more reliable than the duplex counterpart. Furthermore, the use of triplex clad decreased the conditional core damage probability in the reference LBLOCA scenario by three orders of magnitude. These results were caused by redundancies in clad functions carried out by different layers in the triplex structure.

Stress analysis demonstrated that the total stress on the cladding converged to the irradiation swelling stress profile during the reference LBLOCA scenario. This swelling stress played a determining role to the SiC clad vulnerability. It caused the highest axial power peaking height to be the most vulnerable axial fuel rod location in the reference LBLOCA scenario. Additionally, it conditioned the duplex clad to be vulnerable to pressure buildup within the fuel rod. Meanwhile, the triplex clad was tolerant to pressure stresses, yet sensitive to thermal shocks experienced due to delayed LOCA reflood phase.

A sensitivity study showed that the safety benefits of SiC clad could be leveraged to relax the actuation timing of ECCS without increasing the plant risk. Several possible ECCS timing densities were presented based on our assumption on the initial ECCS density. These new density profiles showed that the ECCS actuation could be delayed within tens to several hundreds of seconds compared to the initial ECCS timing for the Zr clad. Such a short additional time may provide limited benefits in the LBLOCA accident mitigation procedure. However, it may considerably prolong the start-up time for diesel generators and pumps, which consequently increase the start-up probability of these components. This benefit will be investigated in future research.

5. Acknowledgment

This work is supported by and is performed in conjunction to the United States Department of Energy Federal Grant number DE-NE0008760.

6. References

- [1] U.S. DOE, "Development of Light Water Reactor Fuels with Enhanced Accident Tolerance (Report to Congress)," U.S. DOE, Washington DC, 2015.
- [2] Idaho National Laboratory, "INL/EXT-12-27090: Advanced LWR nuclear fuel cladding system development trade-off study," INL, Idaho Falls, Sept. 2012.
- [3] Y. Lee, H. Kim and H. No, "Failure probabilities of SiC clad fuel during a LOCA in public acceptable simple SMR (PASS)," *Nuclear Engineering and Design*, vol. 292, pp. 1-16, 2015.
- [4] Y. Lee, T. McKrell and M. Kazimi, "Thermal shock fracture of hot silicon carbide immersed in water," *Journal of Nuclear Materials*, vol. 467, pp. 172-180, 2015.
- [5] Y. Lee, H. No and J. Lee, "Design optimization of multi-layer Silicon Carbide cladding for light water reactors," *Nuclear Engineering and Design*, vol. 311, pp. 213-223, 2017.
- [6] Y. Lee and M. Kazimi, "A Structural Model for Multi-Layered Ceramic Cylinders and Its Application to Silicon Carbide Cladding of Light Water Reactor Fuel," *Journal of Nuclear Materials*, vol. 458, pp. 87-105, 2015.
- [7] Y. Lee, "Safety of Light Water Reactor Fuel with Silicon Carbide Cladding (Doctoral Dissertation)," Department of Nuclear Science and Engineering, Massachusetts Institute of Technology, Cambridge, 2013.
- [8] T. Zhang, R. Yue, X. Wang and Z. Hao, "Failure probability analysis and design comparison of multi-layered SiC-based fuel cladding in PWRs," *Nuclear Engineering and Design*, vol. 330, pp. 463-479, 2018.

- [9] Y. Deng, K. Shirvan, Y. Wu and G. Su, "Probabilistic View of SiC/SiC Composite Cladding Failure Based on Full Core Thermo-Mechanical Response," *Journal of Nuclear Materials*, vol. 507, pp. 24-37, 2018.
- [10] G. Singh, K. Terrani and Y. Katoh, "Thermo-mechanical assessment of full SiC/SiC composite cladding for LWR applications with sensitivity analysis," *Journal of Nuclear Materials*, vol. 499, pp. 126-143, 2018.
- [11] V. A. Avincola, P. Guenoun and K. Shirvan, "Mechanical performance of SiC three-layer cladding in PWRs," *Nuclear Engineering and Design*, vol. 310, pp. 280-294, 2016.
- [12] W. Li and K. Shirvan, "ABAQUS analysis of the SiC cladding fuel rod behavior under PWR normal operation conditions," *Journal of Nuclear Materials*, vol. 515, pp. 14-27, 2019.
- [13] Y. Lee, B. Kim and H. No, "Improving Safety Margin of LWRs by Rethinking the Emergency Core Cooling System Criteria and Safety System Capacity," *Nuclear Engineering and Design*, vol. 307, pp. 402-410, 2016.
- [14] D. Mandelli, C. Parisi, N. Anderson, Z. Ma and H. Zhang, "Accident tolerant fuels: A PRA comparison," in *ANS PSA 2019*, Charleston, 2019.
- [15] Idaho National Laboratory, "Plant-level scenario-based risk analysis for enhanced resilient PWR - SBO and LBLOCA (INL/EXT-18-51436)," INL, Idaho Falls, 2018.
- [16] Oak Ridge National Laboratory, "Emergency core-cooling systems for light-water-cooled power reactors (ORNL-NSIC-24)," ORNL, Oak Ridge, Oct. 1968.
- [17] G. Petrangeli, "Chapter 4: The classification of accidents and a discussion of some examples," in *Nuclear Safety*, Burlington, Butterworth-Heinemann, 2006, pp. 35-52.
- [18] H. Chung, "Fuel Behavior Under Loss-of-Coolant Accident Situations," *Nuclear Engineering and Technology*, vol. 37, no. 4, pp. 327-362, 2005.
- [19] C. Frepoli, J. Yurko, R. H. Szilard, C. Smith, R. Youngblood and H. Zhang, "10 CFR 50.46c Rulemaking: A Novel Approach in Restating the LOCA Problem for PWRs," *Nuclear Technology*, vol. 196, no. 2, pp. 187-197, 2017.
- [20] U.S. Nuclear Regulatory Commission, "Regulatory Guide 1.174 Rev. 3: An approach for using probabilistic risk assessment in risk-informed decisions on plant-specific changes to the licensing basis," US NRC, Washington DC, 2018.
- [21] Information Systems Laboratories, Inc., "RELAP5/MOD3.3 Code Manual Volume I: Code Structure, System Models, and Solution Methods," Information Systems Laboratories, Inc., Idaho Falls, 2016.
- [22] T. Sawabe, M. Akiyoshi, K. Yoshida and T. Yano, "Estimation of neutron-irradiation-induced defect in 3C-SiC from change in XRD peak shift and DFT study," *Journal of Nuclear Materials*, vol. 417, no. 1-3, pp. 430-434, 2011.
- [23] Y. Katoh, T. Nozawa, L. Snead, K. Ozawa and H. Tanigawa, "Stability of SiC and its composites at high neutron fluence," *Journal of Nuclear Materials*, vol. 417, no. 1-3, pp. 400-405, 2011.
- [24] A. Mieloszyk, "Assessing Thermo-Mechanical Performance of ThO₂ and SiC Clad Light Water Reactor Fuel Rods with a Modular Simulation Tool (Doctoral Dissertation)," Department of Nuclear Science and Engineering, Massachusetts Institute of Technology, Massachusetts, 2015.
- [25] Oak Ridge National Laboratory, "ORNL/TM-2017/385: SiC/SiC Cladding Materials Properties Handbook," ORNL, Oak Ridge, Aug. 2017.
- [26] D. Fynan, "Uncertainty Quantification for Reactor Safety Analysis (Doctoral Dissertation)," Nuclear Engineering and Radiological Sciences, University of Michigan, Ann Arbor, 2014.
- [27] Oak Ridge National Laboratory, "Handbook of LWR SiC/SiC Cladding Properties Revision 1 (ORNL/TM-2018/912)," ORNL, Oak Ridge, 2018.

- [28] Oak Ridge National Laboratory, "Radial Thermal Conductivity Measurement for SiC Composite Tubes (ORNL/TM-2018/1007)," ORNL, Oak Ridge, 2018.
- [29] L. Snead, T. Nozawa, Y. Katoh, T. Byun, S. Kondo and D. Petti, "Handbook of SiC properties for fuel performance modeling," *Journal of Nuclear Materials*, vol. 371, no. 1-3, pp. 329-377, 2007.
- [30] Y. Deng, K. Shirvan, Y. Wu and G. Su, "Probabilistic View of SiC/SiC Composite Cladding Failure based on Full Core Thermo-Mechanical Response," *Journal of Nuclear Materials*, vol. 507, pp. 24-37, 2018.
- [31] Oak Ridge National Laboratory, "Interlaboratory Round Robin Axial Tensile Testing of Tubular SiC/SiC Specimens (ORNL/SR-2017/397)," ORNL, Oak Ridge, 2017.
- [32] Oak Ridge National Laboratory, "Emergency Core Cooling Systems for Light Water Cooled Power Reactors (ORNL-NSIC-24)," ORNL, Oak Ridge, 1968.
- [33] M. Idris, S. Yamazaki, K. Yoshida and T. Yano, "Recovery behavior of high purity cubic SiC polycrystals by post-irradiation annealing up to 1673 K after low temperature neutron irradiation," *Journal of Nuclear Materials*, vol. 465, pp. 814-819, 2015.
- [34] M. Idris, S. Yamazaki, K. Yoshida and T. Yano, "Recovery behavior of SiCf/SiC composites by post-irradiation annealing up to of 1673 K," *Energy Procedia*, vol. 131, pp. 413-419, 2017.
- [35] D. Fynan and K. Ahn, "Implicit treatment of technical specification and thermal hydraulic parameter uncertainties in gaussian process model to estimate safety margin," *Nuclear Engineering and Technology*, vol. 48, pp. 684-701, 2016.